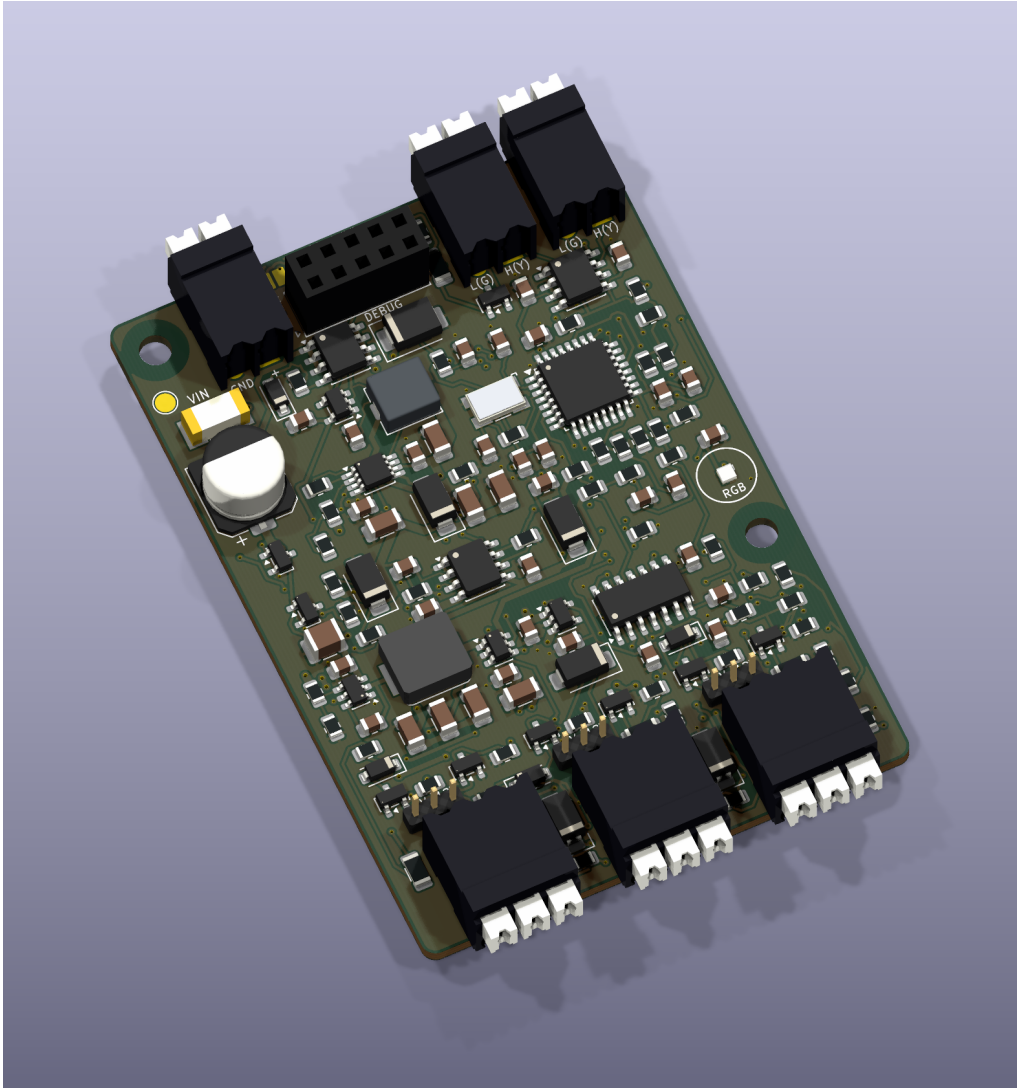
FRC
Sensor CAN

Revision H SC12 · Prototype engineering and assembly guide

Three industrial / logic sensor inputs
Regulated 12 V and 5 V sensor power



2026-09-09

Rev H adds RGB diagnostics and inexpensive header programming while retaining protected sensor power and physical voltage jumpers. No physical prototype has been tested. This guide accompanies the engineering evidence and defines first-article acceptance.

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1 Product overview

Source: README.md

Three digital sensor inputs over CAN, with regulated **5 V / 12 V sensor power** and software-selectable **NPN, PNP, push-pull, floating detection and keeper**. Removable 2 mm jumpers select each port's power voltage. All components mount on the top.

Rev H adds one enclosure-facing RGB status LED and a hand-soldered recessed female programming socket for a ten-board batch. The LED is on the right, above the mounting hole, with a 6 mm circular area reserved for a printed light duct; see the enclosure dimensions. It retains Rev G's power protection and physical voltage jumpers. The 12 V reservoir is upstream of a controlled output eFuse; both rails have default-off enables and CAN fault reset. Two ADC channels measure the protected rails. Engineering review · verification and bench procedure. This design uses its own CAN protocol; it is not CTRE Phoenix or GrappleHook compatible.

Rev H status: RGB/header hardware, firmware and focused checks are complete. The extra full power-transient rerun was stopped before completion; it is not claimed as passed. See the check summary.

Specification	Rev H
Board supply	6-22 V DC
Sensor supplies	Regulated 5 V / 12 V; **300 mA shared per rail**, not per port
Signal targets	Positive 0-30 V; LOW ≤1.2 V, HIGH ≥2.4 V; common ground
Input setup	Software bias and active-state selection independently per channel
Protection	Reverse input, fuse/TVS, separate protected 5 V and 12 V outputs, controlled 12 V inrush
Fault reset	Remove the fault, then use CAN power reset; hardware limits remain independent
Controller	STM32F042K6T6, TCAN3413; 1 Mbit/s classic CAN
Software	App 0.9.0, bootloader 1.1.0, hardware 8, protocol 4
Servicing	CAN discovery, ID/settings and recoverable firmware upload; initial installation through a 2.54 mm SWD service socket
PCB / assembly	**52 × 74 mm**, four copper layers, 130 top SMT parts + ten hand-soldered parts + three loose caps

Parts purchasing is about **$28.99 per board at ten boards**, including supplier minimum packs, hand-fitted parts and three loose caps. RGB/header changes add about $0.18 in parts versus G; the design remains about $6.15 below E. PCB, assembly, freight and tax are extra. Cost calculation.

Change voltage jumpers only with power disconnected. A missing cap disconnects port power, not its signal wire. Input mode changes use CAN software. Rail voltage readback uses the MCU ADC and factory voltage-reference calibration. External signal inputs remain digital; rail measurement does not add user analog-input or encoder support.

The current layout has rounded corners, aligned input-bias transistors, a recessed female debug socket, improved edge clearances and visible voltage-selector/input labels. It remains 52 × 74 mm with top-side assembly. See connector and layout details.

2 Pre-order decision

Source: docs/ORDER_READINESS.md

Rev H retains Rev G's power-path repairs and adds RGB diagnostics plus a recessed female service socket. The LED/header checks pass; the additional full power-transient rerun was stopped before completion and is not reported as a Rev H pass. No physical board has been tested; the package is a prototype design with reproducible engineering evidence.

The latest clearance, socket, rounded-outline and alignment edits receive CAD, pinout and manufacturing cross-checks only. Final simulations are deferred until the design is finalized, as requested. Sensor-rail distribution was rerouted for the moved jumpers and local transistor clearances; the retained DC report is from an earlier layout and is not presented as a fresh pass.

The power-stage simulation findings below refer to the completed Rev G baseline. Current H evidence separately covers the added RGB load allowance, native geometry, pin mapping and firmware.

Changes that address the earlier failures

Rev F finding	Retained power implementation	Verification
Input-side limiter interrupts loaded boost startup	U8 connects to internal 5 V; TPS259620 output eFuse connects the sensor load after the raw boost reservoir	Combined low-input startup with 66 µF raw reservoir, 100 µF external load and 300 mA load
Reservoir discharge passes through the boost short-current path	U9 limits the protected output; C62-C64 remain upstream; D5 clamps negative output excursions	Running short, cold short, sustained fault and restart scenarios
Firmware cannot reset power protection	PA15 controls U4; PB7 controls Q3/Q2 and U9 enable	Register-level pin tests, application sequencing and addressed CAN reset tests
Reset/unpowered MCU needs a safe output state	R71/R75 pull enable signals LOW; R70 derives the 12 V inhibit from raw BOOST_12V	Native pin/net checks and default-off circuit review
Coarse rail-presence indication misses undervoltage	Separate 11:1 dividers and RC filters feed PB0/PB1 ADC; firmware uses calibrated internal reference	ADC register tests, rail-state tests and divider/current budgets
Local boost bypass below recommended value	C61 increased to 1 µF; 22 µF bulk C60 retained	Exact-MPN BOM/model/parity checks
Buck input bypass left remote from U1	C2/C3 moved beside VIN; explicit local top-side ground return, with bootstrap routing clear of it	Native path checks require C3→VIN below 2.2 mm and C2→VIN below 5.5 mm; fresh DRC and Gerber connectivity

The 5 V/12 V selection remains a removable 2 mm cap on each three-pin header. All 140 fitted parts remain on top: 130 SMT factory placements plus ten user-fitted through-hole parts. The PCB is 52 × 74 mm, four copper layers.

Evidence to read

Release check index, electrical budgets, native-route DC calculation and pin review, 768-corner averaged boost-loop screen, and vendor-model transient cases with earlier complete Rev G evidence.

The DC calculation uses nominal 35 µm copper at 85°C and 20 µm via plating, with ideal pad contacts. It checks the 120 mV boost-input PCB allowance at a conservative 1.3 A boost branch, 510 mA 5 V branch, 130 mA logic load and 20 mA RGB/buffer load. Ground return, connectors, cables, actual copper tolerances and temperature rise need measurement.

Pin expectations are checked independently against manufacturer functions for the MCU, CAN transceiver, comparators and power switches. Native ERC/DRC, model/schematic/PCB parity and reconstructed Gerber connectivity check the actual delivered design rather than only the conceptual schematic.

Limits and first-article acceptance

The vendor transient model uses an idealized upstream 5 V source; it does not simulate the LMR51420 loop, robot wiring, all silicon corners or an assembled board. The eFuse's floating OVCSEL setting is a roughly 13.8 V fault clamp, not a precision 13 V trip. Software rail monitoring supplements the independent hardware current limits. It does not make a fault-proof or reverse-powered output.

Qualify startup, both rails loaded together, the exact sensor models and cables, brownouts, cold/running shorts, temperature and CAN programming/recovery using the validation procedure. Record PCB revision H, exact BOM and firmware identity. Simulation and manufacturing checks reduce the risk of a first order; they cannot guarantee first-board success.

Primary pin/application references: STM32F042, TPS2596, TPS61085, TPS2553, TCAN3413, LM339LV.

The RGB load is included in the increased 0.15 A internal logic/indicator reserve. Check D10 colors, timer waveform and reset/handoff behavior on the first physical board. The recessed J8 socket needs five male-ended mapped SWD leads; see status and programming.

3 Power and input circuit

Source: DESIGN_PLAN.md

Rev H retains G's protected power circuit and adds RGB status and ordinary-header programming. The regression report records every change against the frozen G circuit.

Supply path

```
6-22 V → fuse / reverse protection / TVS → internal 5 V, 2 A
                                          ├─ 3.3 V MCU / CAN / comparators
                                          ├─ U4 latch limiter → SENSOR_5V
                                          └─ U8 boost → reservoir → U9 eFuse → SENSOR_12V
SENSOR_5V / SENSOR_12V → removable jumper per port → sensor power
```

U1 remains LMR51420XDDCR. U8 remains TPS61085DGKR, with the 3.3 µH shielded inductor, B240A diode and 13 kΩ / 33 nF compensation. C61 is now a 1 µF local bypass. C60 remains 22 µF, and C62-C64 remain three 22 µF reservoir capacitors. Their effective capacitance at 12 V is much smaller than their nameplate value; the checks include both explicit derated and nameplate cases.

A boost has a direct path through its inductor and diode. Rev H puts **TPS259620DDAR after the reservoir**, so a sensor short cannot discharge that reservoir directly through the port. R62=2.61 kΩ sets about 357 mA nominal output current limit; the tolerance budget is in electrical_checks.json. Useful sensor load remains **300 mA shared per rail**. C70=47 nF controls output slew; C65/C71 provide local input/output bypassing. D5 clamps negative output transients and D6 limits raw-rail transients. These parts do not authorize reverse powering a sensor port.

U9's UVLO divider requires approximately 11.1 V on the raw boost before connection. OVCSEL is intentionally open for its **13.8 V-class fault clamp**, not a precision 12 V ceiling. R70 pulls Q2's gate from raw 12 V, holding UVLO low by default. PB7 HIGH enables Q3, which removes that inhibit. R75 holds Q3 off without MCU power. U4's PA15 enable has a hardware pulldown. Both branches therefore start disabled; their current limits operate independently of firmware.

Firmware starts 5 V after 10 ms and 12 V after 60 ms. It allows 200 ms for startup, requires 100 ms healthy voltage before counting inputs, and latches a running low-voltage fault after 20 ms. Hardware fault indications or excessive measured voltage disable the affected rail. Removing a short does not automatically restart it: use the explicit CAN power-reset command. The bootloader keeps both rails off.

Rail measurement and signals

PB0/ADC8 and PB1/ADC9 measure the protected 5 V and 12 V rails. Each divider uses two 100 kΩ precision series resistors, a 20 kΩ bottom resistor and a 10 nF filter. Firmware uses 239.5-cycle acquisition and the factory VREFINT calibration value. Readback and rail states are available through CAN service. This measures the board rails, not the voltage at a remote sensor.

The former coarse fourth comparator is unused, with defined inputs and an unconnected output. The three signal comparators and electronic NPN/PNP/push-pull/float/keeper circuits retain their original values. Two pulldown controls move to PB5/PB6; channel 3 stays on PB4. Selecting 12 V sensor power never changes the signal pull-up from 3.3 V.

External inputs remain digital: target LOW ≤1.2 V, HIGH ≥2.4 V, positive 0-30 V and common ground. Sensor off-state leakage, cable capacitance and residual voltage must meet the input-mode limits. Rail ADCs do not add external analog or encoder support.

Layout and assembly

The board is 52 × 74 mm with four copper layers and all 140 fitted components on top. The factory places 130 SMT parts; the user fits six terminals, three voltage headers and one J8 service header, then three loose caps. U9 uses a split-paste exposed pad and surrounding thermal vias. No blind/buried vias or second-side assembly is required.

One inner layer provides continuous ground; the other also carries a few slow control/service connections. Switching, feedback, compensation and crystal routes remain on top. Native and independently reconstructed fabrication geometry are checked separately.

Use hardware 8, application 0.9.0, bootloader 1.1.0 and protocol 4 from the same release. Verification distinguishes calculations and simulation from physical qualification. Cost basis includes supplier minimums and excludes assembly/freight/tax.

Rev H design envelope: at most 100 µF total external capacitance on the shared 12 V rail, and 22 µF on the shared 5 V rail, at 300 mA per rail. The startup calculation includes +20% capacitance and the current-limit minimum; the 470 µF simulation is a typical diagnostic beyond that envelope. Qualify nonlinear sensor inrush separately.

RGB diagnostics and low-volume programming

H replaces the single green LED with one SK6805-EC15, a 5 V AHCT buffer, data pulldown, series resistor and local bypass capacitors. PB3 remains the status pin and uses timer/DMA serialization. J8 is a hand-fitted recessed female 2×5 2.54 mm socket with the same electrical pin numbers and odd/even physical pattern as the preceding Rev H header. Status, case opening and inexpensive programmer.

4 Wiring and programming header

Source: docs/WIRING.md

Use the included hardware-8/protocol-v4 firmware and tools; older-revision firmware must not be used on Rev H. Board coordinates below are viewed from the component side, origin at the top-left PCB corner. Bottom silkscreen is read after turning the board over. Do not infer pin numbering from left/right after flipping.

Connector	Pin 1	Pin 2	Pin 3
J1 power	Positive input, 6-22 V	GND	-
J2, J3 CAN	CAN H, yellow: H(Y)	CAN L, green: L(G)	-
J4 input 1	Selected sensor power	GND	Signal
J5 input 2	Selected sensor power	GND	Signal
J6 input 3	Selected sensor power	GND	Signal

Each terminal pole has two solder legs; they are one electrical contact. J2/J3 are connected in parallel for bus continuation, with no onboard CAN ground terminal. Common ground comes through robot power. Use twisted H/L wiring; keep branches short. Strip 8 mm using the conductor range specified by Weidmüller for the exact terminal. Press the white release button for removal and for conductor types requiring it. Provide cable strain relief independently of the terminal.

Both CAN connectors are marked **L(G)** for CAN Low / green and **H(Y)** for CAN High / yellow, following WPILib's FRC CAN wiring convention. The four front labels share one horizontal baseline; the underside repeats the same color abbreviations.

Feed J1 from appropriately fused robot distribution. F1 is secondary board protection. Do not backfeed J4-J6 power pins, parallel the two sensor rails, or apply external power to the programming rail. The board is not galvanically isolated.

Sensor power and software bias

The factory leaves JP101/JP201/JP301 unpopulated for hand soldering. Fit the specified 2 mm headers, then leave caps off for initial bring-up. With power disconnected, fit ONE cap across 1-2 for 5 V or 2-3 for 12 V. No cap disconnects port power. Connecting all three power pins shorts the rails.

The headers are centered at X=7, 23 and 39 mm, Y=60.4 mm. Pins 1-2-3 run left to right from the component side; pin 1 is square. Above each header, the front labels show 5V on the left and 12V on the right. V1/V2/V3 are the selected sensor-power outputs, GND is the return, and IN1/IN2/IN3 are signal inputs. There are no physical input-bias selectors. Set each channel to NPN, PNP, push-pull, float detection or keeper in software; see input modes.

Use Harwin M22-2010305 headers and Sullins SPN02SVEN-RC closed-top caps (2 mm pitch; not 2.54 mm). Caps are friction-retained, not mechanically latched. After voltage/functional verification, support retention with a small removable bead of electronics-grade neutral-cure silicone between the cap exterior and header body, clear of the contacts. This is a proposed retention method; qualify it for your robot vibration/impact environment and enclosure. Allow full cure before use and inspect seating during service. Do not claim these are vibration-qualified merely because there is no slider. Bridge JP2 only at a CAN bus end that needs a 120 Ω termination. Leave it open at intermediate nodes; normally there are two terminations on the entire bus.

Examples: select `npn` for a sinking output (default active LOW), `pnp` for a sourcing output (active HIGH), or `push-pull` for an actively driven logic signal. NPN pull-up current is at most approximately 3.5 mA. Both NPN and PNP compatibility budgets allow 100 µA OFF leakage. No onboard 3.3 V sensor-power output is provided; the signal pull-up is not a sensor supply.

Design targets: 0-30 V positive input, LOW ≤1.2 V / HIGH ≥2.4 V, common-ground offset budget ±0.1 V. Do not use as an analog input, quadrature encoder, IO-Link port or continuous negative-voltage input. Sampling is 1 ms with 1-100 ms debounce; short pulses can be missed. Sensor power budgets are shared, 300 mA per rail. U4 independently limits 5 V; U9 limits the protected 12 V output at about 357 mA nominal (roughly 314-401 mA tolerance screen). Firmware latches a failed rail off and does not retry automatically. Remove the

fault, then issue addressed CAN `power-reset` for the affected rail, or cycle board power. ADC dividers measure both rails; CAN service reports millivolts and enable/good/fault states. These are supply measurements, not sensor analog inputs or current measurements. A fault on either rail makes the health-checked driver input unavailable. See protocol.

Representative industrial sensors

Regulated 12 V is a suitable nominal supply for the following families, subject to exact model and cable checks. Connect the selected output only; insulate unused complementary output wires separately. Brown/blue/black is common supply/ground/output coloring, but follow the specific datasheet rather than assuming colors.

Family	Published limits	Rev H setup / limitation
Omron E3Z three-wire NPN/PNP	12-24 V ±10%; typically 30 mA maximum, through-beam pair 35 mA; residual ≤1 V below 10 mA	Power 2-3 for 12 V. NPN or PNP software mode. Residual voltage fits the LOW target at this board's small load
Banner QS18 standard opposed/diffuse/retroreflective variants	10-30 V; <25 mA excluding output load; standard outputs <1 V at 10 mA	12 V supply is in range. Select NPN or PNP bias to match model; use one output. Account for its 100 ms power-up delay
Banner QS18 DVS/DL/FF variants	NPN residual specified up to 1.5 V at 10 mA	This exceeds the board's design LOW target. Prefer a compatible PNP variant or qualify actual NPN residual at the ~3.5 mA pullup load; do not assume all QS18 NPN models work

Sources: Omron E3Z specifications, Banner QS18 manual. PNP OFF leakage must also meet the input pulldown budget. Observe manufacturer cable/fusing requirements. Two-wire current-loop, AC, analog, IO-Link and safety/OSSD sensors are not supported as interchangeable three-wire inputs.

12 V is specified at the board, not guaranteed at the remote cable end under every transient. The analytical DC divider/reference budget is approximately 11.64-12.38 V before dynamic/distribution allowances; check ≥10.8 V at an E3Z during startup and worst battery/load conditions. Three 30 mA sensors are comfortably within the 300 mA shared design target, but include emitters, indicator loads, cable losses and inrush. The board's 100 ms rail qualification is not proof that every sensor has completed its own startup.

Programming header

J8 is a recessed female **2×5, 2.54 mm through-hole socket**, centered at (24, 6.30) mm. Its electrical mapping is custom, not an ARM/STDC ribbon pinout. Use male-ended mapped leads or a correctly wired 2×5 male plug. In the component-side view, the upper row is **2, 4, 6, 8, 10** left to right at Y=5.03 mm; the lower row is **1, 3, 5, 7, 9** at Y=7.57 mm. Pin 1 is square at (18.92, 7.57) mm. Hole diameter is 1.10 mm. The ten electrical pin assignments and hole positions are unchanged from the previous male header.

J8 pin	Signal	Connection
1	NRST	Probe reset; connect for factory programming
2	MCU UART RX	3.3 V UART adapter TX
3	MCU UART TX	UART adapter RX
4	SWCLK	SWD clock
5	SWDIO	SWD data
6	+3V3	Probe VTref sense **only**, board output
7	GND	Probe ground
8	GND	UART ground
9	BOOT0	Normally leave open; onboard pulldown

J8 pin	Signal	Connection
10	GND	Ground

Power the board through J1 during programming. Leave adapter/probe power disconnected; pin 6 is a voltage-reference input to the probe, not a power input to SC12. Leave BOOT0 open. One reusable ST-Link and five mapped wires program all ten boards. See status and low-cost programming for the exact probe connections and LED legend. Remove programming wires for robot operation.

Disabling a rail opens its supply switch; it does not actively discharge the output. Stored charge decays through connected loads and the voltage divider. `power-reset` clears the protection latch and re-enables the rail; it does not guarantee a particular power-off voltage or reset interval for every sensor.

Rev H design envelope: at most 100 µF total external capacitance on the shared 12 V rail, and 22 µF on the shared 5 V rail, at 300 mA per rail. The startup calculation includes +20% capacitance and the current-limit minimum; the 470 µF simulation is a typical diagnostic beyond that envelope. Qualify nonlinear sensor inrush separately.

5 Software input modes

Source: docs/INPUT_MODES.md

Each input has its own electronically controlled pull-up and pull-down; there are no input-bias jumpers or solder selectors. Select the mode in the desktop setup tool, CAN service CLI, or Java/C++ runtime API. This is an independent design with CANdi-like input choices; CTRE Phoenix and GrappleHook are not compatible protocols.

Mode in setup tool	Electrical behavior	Default active condition	Typical use
`npn`	Pull toward 3.3 V through a diode and 1 kΩ	LOW	NPN / open collector sinking output
`pnp`	Pull toward ground through 10 kΩ	HIGH	PNP sourcing output
`push-pull`	Both bias switches off	HIGH	Actively driven 3.3 V, 5 V or industrial logic
`float`	Alternate pull-up/down to classify the source	DRIVEN	Slow contact / high-impedance diagnostics
`keeper`	Bias toward the last sampled level; starts LOW	HIGH	Hold a released line near its last logic level

Active condition is independently selectable as `high`, `low`, `driven`, `floating`, `not-high`, or `not-low`. Inversion is applied afterward, then debounce. Normally leave the inversion mask at zero when using these active conditions. Firmware defaults to push-pull/HIGH on all channels; it does not guess a sensor type.

Wiring and electrical limits

Power, ground and signal remain the three sensor-terminal contacts. Choose 5 V or 12 V sensor power with the separate removable power cap, with board power disconnected. These power caps and CAN termination are not input-bias settings. Keep a common signal ground.

The pull-up uses the internal **3.3 V rail regardless of sensor supply**. Its series diode blocks most current from a 5-30 V signal toward the MCU rail. The voltage-rated pull-down MOSFET switches a 10 kΩ resistor. The LM339LV comparator, protected divider and hysteresis still receive the field signal; no field-voltage connection goes directly to an STM32 GPIO. This remains digital sensing, not analog acquisition.

Targets: positive 0-30 V input, LOW ≤1.2 V, HIGH ≥2.4 V, ±0.1 V ground-offset allowance. The compatibility budget allows 100 µA OFF leakage for NPN or PNP sensors. NPN LOW residual must remain ≤1.2 V at up to approximately 3.5 mA sink current. Verify the exact sensor datasheet. A nominal 5 V push-pull signal works with bias off. A 3.3 V open-drain output may use NPN mode if it tolerates the board's pull-up while the sensor is unpowered; otherwise use an external pull-up to its own supply with onboard bias off.

Both switches default off through external gate resistors during reset and in the bootloader. Firmware applies saved modes during startup and suppresses measurements until input/rail settling completes. Removing a power cap does not disconnect SIGNAL; an enabled pull-up can inject current into an unpowered sensor. Set push-pull/bias-off before servicing an externally connected unpowered device. Reverse diode leakage is finite: the 10 kΩ 3.3 V rail bleed budgets about 0.61 V from the three bias diodes alone using the BAT46W 60°C specification. Charged sensor-rail capacitors can temporarily add current through the ADC dividers. This is not isolation or a whole-board unpowered-voltage guarantee.

Floating detection and counts

Float mode waits at least 2 ms after each bias change. Two matching full pull-up/down pairs are required, so initial classification needs approximately 8 ms, plus configured debounce for subsequent logical changes. Input-level telemetry reports HIGH, LOW, FLOATING or UNKNOWN (unsettled). Drivers mark logical inputs unavailable while any channel is unsettled or either rail is unhealthy.

Probe transitions never enter the logical edge counter. Counters track debounced inactive-to-active transitions. They pause during rail faults and maintenance; startup/reconfiguration re-baselines without

inventing an edge. Use float mode for slow signals, and qualify the real cable capacitance, noise and sensor leakage. Fast changes during a probe pair can be misclassified; fixed NPN/PNP/push-pull modes are preferable for ordinary sensing. This is not an encoder interface.

High impedance can mean a disconnected cable, an open contact, or the normal inactive state of an NPN/PNP output. Float detection cannot distinguish these causes. Keeper mode biases a released wire; it does not retain a value across reset or report a disconnection.

Save an example configuration

From `current/`, after discovery and physical identification:

```
python host/can_service.py --bridge localhost:5810 configure \
  --uid UID --device 17 --invert 0 --debounce 2 \
  --modes npn,pnp,push-pull
```

Replace `UID` with the discovered 24-digit hexadecimal serial. The tool stages all three settings, commits with the ID/settings record, resets, and verifies persistent readback. If `--modes` is omitted, existing saved input settings are preserved. Optional `--active low,high,high` overrides the defaults. The desktop window offers the same choices independently for each channel.

Java `configureInputsRuntime(InputMode[], ActiveWhen[])` and C++ `ConfigureInputsRuntime(...)` apply volatile settings. Read back `inputState()` / `ReadInputState()` to verify the configuration word and classification; runtime CAN commands have no acknowledgment and do not write flash. Use CAN service to persist. See wire format, service and drivers.

Engineering checks and simulations do not qualify electrical behavior. Bench-test all modes across supply, temperature, cable length, hot-plug, sensor startup and robot vibration before deployment. See validation.

6 Status and programming

Source: docs/STATUS_AND_PROGRAMMING.md

Rev H has one RGB LED, D10, in the open area on the right, above the right-hand mounting hole. A silkscreen circle and RGB label identify it. It runs from the internal 5 V supply, so it can indicate a sensor-power fault while the protected outputs are off. Sensor-voltage selection remains on the removable 2 mm jumpers.

Reading the indicator

Pattern	Meaning
Green, one short pulse per second	Configured, rails qualified, telemetry enabled, and a CAN transmission acknowledged within 500 ms
Amber, one short pulse per second	Waiting: unconfigured, startup, maintenance, or telemetry disabled; also the initial CAN connection grace period
Red, two short pulses per second	Latched sensor-power fault or clock-start failure; inspect CAN diagnostics and rail measurements
Red, equal on/off every 250 ms	CAN initialization/error-passive/bus-off fault, or no successful transmission after the initial one-second grace period
Blue, one 200 ms pulse per second	Bootloader, including CAN update/recovery
Magenta, rapid 50 ms on/off	Three-second identify request; application power/CAN faults take priority
Dark or frozen color	Check board power, reset and firmware; this alone does not prove the outputs are off

A CAN acknowledgment proves that another node accepted the frame at the bus level; it does not prove the robot application consumed it. Green does not certify sensor type, jumper voltage, robot enable state or healthy external wiring. There are no per-input LEDs. This is SC12's own status scheme and CAN protocol.

Enclosure location

Coordinates use the component-side view, top-left PCB corner as (0,0). D10 is centered at **X=48 mm, Y=39.5 mm**: 4 mm from the right edge and 7 mm above mounting hole H2. Its nominal body is 1.5 × 1.5 × 0.65 mm. A **6 mm diameter area** centered on D10 is reserved against other component courtyards for a future circular light duct. C75 sits just above this area and remains connected by a short local supply trace.

For a first printed enclosure, use a duct with **6 mm maximum outside diameter and 4 mm inside diameter**. These are nominal starting dimensions, not a tolerance-qualified enclosure. The duct must hang from the case, without loading the PCB or LED. Start its lower face **0.2-0.5 mm above the LED**, or nominally **0.85-1.15 mm above the top PCB surface**. Maintain the separate mounting-hardware clearance at H2; do not combine the duct with a screw boss.

The LED should be visible from above, but brightness and viewing angle have not been measured on hardware. Keep a hollow duct short and its opening unobstructed; a translucent window or light-pipe insert can make the case indicator easier to see from an angle. Check all RGB patterns through the actual print under workshop and competition lighting before finalizing the case. The CAD render shows the package without illumination.

J8 is centered at (24,6.30) mm, with a 13.10 × 5.00 mm insulator and approximately 8.5 mm total height above the PCB. Its tails extend about 1.6 mm beyond a 1.6 mm PCB. Allow access through a removable lid and space for plugged leads during servicing. The completed STEP model and mechanical drawing include both parts. All assembly remains on top.

One programmer for the ten-board batch

Hand solder the **XFCN PM254V-12-10P-H85 recessed female 2×5, 2.54 mm J8 socket** included in the hand-solder BOM. Its contacts sit inside the insulating body instead of projecting above it. The estimated undiscounted price is $0.1158 each at ten, approximately $0.044 more per board than the former male header. No special pogo cable is required. Keep one short five-wire harness and reuse it across all boards. Disconnect it for robot use.

Use an existing ST-Link with external-target SWD, reset and voltage sensing, or a Nucleo's detachable/external ST-Link according to that specific board's manual. If buying a programmer, the genuine STLINK-V3MINIE was listed at about **$26 on 2026-09-09**, before shipping/tax and USB cable. That is a one-time bench tool, about $2.60 per board spread over ten, plus a small wire/adapter allowance.

The V3MINIE's CN2 edge pads are **2 mm**, and its supplied ribbon is **1.27 mm STDC14**: neither directly plugs into this 2.54 mm socket. Make one mapped adapter, or solder strain-relieved flywires to CN2 following the probe manual. Verify every connection unpowered. Use ordinary male 2.54 mm leads at the SC12 socket end. A single 2×5 male plug with only the required wires populated is easier to orient consistently; match the pin-one triangle and the custom mapping.

Signal	SC12 J8	STLINK-V3MINIE CN2 edge pad	Alternative CN4 STDC14 pin
NRST	1	9	12
SWCLK	4	4	6
SWDIO	5	3	4
3.3 V target reference, sense only	6	10	3
GND	7	6	5

Use only one probe connector path. CN2 uses pads on both faces; identify positions from ST UM2910, connector tables and board views. J8's full UART/BOOT0 mapping and odd/even orientation are in WIRING.md. J8 is a custom signal assignment, not an ARM ribbon-compatible socket.

Power SC12 through J1, with sensor jumper caps removed for first bring-up. Connect J8.6 only to target-voltage sensing; never feed 3.3 V or 5 V from the programmer into it. Leave J8.9 BOOT0 open. With short leads, start SWD at 1 MHz or lower:

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -c 'adapter speed 1000' -f tools/program_factory.tcl
```

Run from `current/`. The script programs the matching combined factory image, verifies it, and sets/verifies bootloader write protection. Power cycle and follow the fixture checks. The first installation needs SWD; later application updates, CAN ID and sensor modes use CAN service. Bootloader or option-byte recovery still requires SWD. USB-C is needed on the external programmer, not on each SC12 board.

Circuit and verification basis

PB3 drives an SN74AHCT1G125 buffer, which converts 3.3 V data to the LED's 5 V logic domain. A 10 kΩ pulldown holds its input low on reset; a 330 Ω series resistor and local 100 nF capacitors support the short board-level link. OPSCO SK6805-EC15 uses GRB data at 800 kbit/s. The buffer is necessary because this LED's minimum HIGH level is 0.78 of its supply; directly driving it with 3.3 V would lack guaranteed margin.

The timer/DMA implementation preserves foreground input sampling and CAN service. Compiled firmware tests exercise the emitted data, timing registers, transfer errors, CAN acknowledgment tracking and application status states. They do not measure real LED waveforms, optical brightness, or reset behavior on silicon. Bench-check colors, timing, CAN unplug/recovery, rail faults and boot-to-application transitions before robot use. The LED may hold its last color until the next valid word if the MCU resets or stops.

Sources: OPSCO SK6805-EC15 A/1 datasheet, TI SN74AHCT1G125, STM32F042 datasheet, ST RM0091.

7 Assembly

Source: docs/ASSEMBLY.md

Factory: fit only the 130 SMT references in Factory_BOM.csv using CPL_JLCPCB.csv. User: buy the ten through-hole parts and three closed-top caps.

Use the exact BOM, assembly drawing, interactive BOM and placement files. Reference labels are in these drawings; dense placement does not leave room to print every reference legibly on the actual top silkscreen.

1. Order the 52 × 74 mm, four-layer, 1.6 mm board with 1 oz finished copper target. Use the exported Gerbers and separate plated/nonplated drill files. Have the fabricator review its actual stackup, annular rings, mask registration and minimum features. ENIG is retained for the fine-pitch SMT pads; hand solder the service header after reflow.

2. Check bare-board continuity and orientation. Apply paste to SMT pads only. A 0.10 mm stencil is a starting point; the assembler must review release and the VSSOP pad apertures. Through-hole header pads, the termination solder bridge and the test point deliberately have no paste.

3. Place 0805/1206/1210 passives, diodes and ICs. Verify C1 polarity, TVS/diode cathodes, pin 1 on each IC and the exact latch-off TPS2553DBVR-1 suffix. U8 TPS61085DGKR uses the supplied 0.65 mm-pitch VSSOP footprint. Confirm D4 cathode orientation and U4 TPS2553DBVR-1 and U9 TPS259620DDAR, including the exposed ground pad and six adjacent thermal vias. D10 must be OPSCO SK6805-EC15 with the supplied pin mapping; U10 is SN74AHCT1G125DBVR. Do not substitute a different addressable LED footprint. There are no components on the bottom.

4. Inspect joints and measure for supply shorts. Hand solder the six horizontal Weidmüller terminals after SMT reflow as the user assembly step. Fit J8, the recessed female 2×5 2.54 mm service socket, with pin 1 at the square pad. Then fit JP101/JP201/JP301: 2 mm headers on the top side, straight and fully seated, with pin 1 at the square pad. Use a removable cap temporarily as an alignment aid with the board unpowered; remove it afterward. Fill both plated holes per terminal pole. Support the board without loading fresh SMT joints. The terminal pins extend below the board.

5. Leave all selectors open for initial bring-up. Supply J1 from a current-limited bench supply, starting at 12 V and a modest current limit. Confirm input polarity, protected input, approximately 5.07 V internal, 3.3 V logic and 12.00 V regulated rails. Sensor headers remain unpowered while power selectors are open.

6. Program the combined factory image through the custom J8 mapping, set/verify boot protection, power cycle and configure a unique CAN ID using CAN service. Follow the fixture procedure. Save each input mode in software and verify readback. Then power off, choose each port's supply and termination settings, inspect one-cap-per-header placement, and check that 5 V and 12 V are not shorted.

7. Perform the qualification checks before connecting a robot harness. Begin with a known load and known logic source, then the actual sensor/cable.

Use M3 hardware at (3.5,4) and (48.5,46.5) mm. Holes are 3.2 mm NPTH. The copper keepout has at least 3.30 mm radius; use hardware whose contact area stays within a 6.4 mm diameter, with insulating washers/spacers as appropriate. Account for component bodies, connector overhang and protruding pins. Do not clamp the PCB between conductive surfaces. The mechanical drawing gives nominal geometry, not a fitted enclosure.

The compact board still benefits from magnification and a fine iron. The retained power stage removes the former QFN and 0402 parts; the MCU and VSSOP still need controlled stencil/reflow assembly. See jumper retention and vibration qualification in WIRING.md.

8 Firmware

Source: firmware/README.md

Bare-metal STM32F042K6T6 firmware for three digital inputs. Build allocation is recorded by the current linker map/build report; stack depth has not been measured. The final two 1 KiB flash pages hold redundant 32-byte versioned configuration with CRC and commit-last writes.

Rev H uses **hardware identity 8** and **protocol 4**. It has a separate image tag and configuration magic so D2 images/settings are rejected. Use the matching drivers and software input modes. The separate power caps select sensor voltage.

CAN service is implemented, physically unqualified: full-UID discovery, identify LED, persistent ID/inversion/debounce and verified uploads. See CAN setup/update, robot bridge and qualification plan.

From `current/`:

```
source tools/env.sh
make -C firmware
bash scripts/test_firmware.sh
```

Outputs: `build/factory.hex` for initial SWD installation; `build/application.sc12` for CAN updates; separate bootloader/application ELF, BIN and MAP files. `sensor_can.hex`/ELF are application-only at 0x08002000. Flash: 8 KiB bootloader, 21 KiB app, 1 KiB manifest, 2 KiB configuration. Both linkers reserve the first 256 SRAM bytes and a 1 KiB minimum stack. The MCU controls six input-bias gates and two sensor-rail enables, and reads divided 5 V/12 V supplies through ADC channels 8/9. Per-input LEDs are not fitted. The 8 MHz external crystal and PLL generate 48 MHz; bxCAN uses 16 time quanta with an 87.5% sample point at 1 Mbit/s. Auto bus-off recovery is enabled; automatic transmit retry is disabled.

Power through J1 from a bench supply. Use the custom 2.54 mm header wiring: VTref sense→6, SWDIO→5, SWCLK→4, ground→7, reset→1. Leave probe power disconnected.

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -c 'adapter speed 1000' \
  -f tools/program_factory.tcl
```

This command has not been exercised on physical Rev H hardware. Verify MCU identity, supply and flash readback. Normal application programming should preserve the two configuration pages; full-chip erase removes them. Version-1/2 configuration is rejected.

UART: adapter TX→J8.2, RX→J8.3, ground→J8.8; 3.3 V, 115200 baud, 8N1; adapter power disconnected.

```
python host/sensor_can.py info --port /dev/ttyUSB0
python host/sensor_can.py configure --port /dev/ttyUSB0 \
  --device 17 --invert 1 --debounce 2
```

Device IDs 0-63 must be unique in this Team Use address space. Invert is a 3-bit mask; debounce is 1-100 ms. UART `SET id invert debounce_ms` and UID-targeted CAN service save persist and reset. Legacy CAN runtime configuration remains volatile. Unconfigured boards accept discovery/service but suppress normal telemetry. Power-cycle after initial SWD programming and verify boot protection before CAN updates. **Both sensor rails default off during reset/programming and remain off in the bootloader.** The application enables 5 V after 10 ms and 12 V after 60 ms; hardware current limiting remains independent of software. On Rev H, leave the caps off JP101/JP201/JP301 to disconnect port power during initial bring-up; configure them only with input power disconnected. Input bias is electronic; CAN termination remains a solder setting. See assembly.

The independent watchdog is serviced only after the full 1 ms foreground cycle. A >20 ms foreground delay, clock failure, hard fault or PVD low-voltage detection puts CAN into standby; the watchdog provides reset recovery. The RGB LED indicates ready (green pulse), waiting/configuration (amber pulse), power/clock fault (red double pulse), CAN failure (red blink), bootloader (blue pulse), or identify (magenta blink). Ready requires a recent successful CAN transmission. See the complete legend and timing. PA3/PA4 read active-low 12 V/5 V fault outputs with internal pullups. PB0/PB1 measure divided 5 V/12 V with the calibrated internal reference. PA15 enables 5 V; PB7 enables the Q3/Q2 12 V stage. PB5/PB6/PB4 drive input pull-downs 1/2/3. A qualified fault disables the rail until explicit CAN reset or a board restart. ADC failure disables both rails. Startup allows 200 ms, then an undervoltage persisting 20 ms latches off; overvoltage or asserted switch fault trips immediately. Healthy rails need 100 ms continuous qualification before counts resume. See `src/power.c` for exact thresholds and unit tests.

Measured health bands are 11.1-12.9 V and 4.55-5.50 V; low-voltage latch thresholds are 10.8 V and 4.4 V. These bands include ADC/reference/divider error and are separate from the calibrated-meter acceptance limits. Overvoltage trips remain 13.0 V and 5.6 V. The electrical report records the measurement-error budget.

Broadcast Disable ID 0 stops telemetry and aborts pending transmissions. Local sampling and application-managed sensor power continue. An explicit `RESUME` UART command or CAN resume command re-enables telemetry; reads never do this automatically. See protocol v4. Host tests cover firmware logic and mocked peripheral boundaries, not actual oscillator startup, electrical timing or STM32 peripheral behavior.

The RGB serializer uses PB3 AF2/TIM2_CH2 and DMA1 channel 2 (TIM2_UP), at 800 kbit/s. It emits GRB, most-significant bit first, with a low reset interval exceeding 200 µs. Timer preloading and two trailing zero slots finish the final bit before DMA stops. The foreground never busy-waits or masks interrupts for LED transmission. A DMA error or transfer lasting beyond 2 ms is stopped and retried later; boot-to-app handoff stops DMA before SRAM is reused. R31 holds the buffer input low during reset. An addressable LED can retain its last color through an MCU reset or crash; interpret repeating patterns, not a frozen color, as firmware activity.

9 CAN protocol

Source: docs/CAN_PROTOCOL.md

1 Mbit/s classic CAN, extended 29-bit identifiers, 8-byte telemetry. This is an original Team Use protocol, not CTRE Phoenix. Device type=10 (Miscellaneous), manufacturer=8 (Team Use), unique device ID=0-63. Address = `(10 << 24) | (8 << 16) | (API << 6) | device`.

The APIs differ from A/B/B-2; Rev C shares these API numbers but uses protocol v2, which Rev H drivers reject. The version and capability checks prevent drivers from interpreting the wrong revision’s payload. Provision each board by full-UID CAN service or 3.3 V UART. An unconfigured board suppresses telemetry but answers service requests; its sensor power hardware is still enabled.

All multibyte fields are little endian. Masks use bits 0-2 for inputs 1-3. Unsupported versions, reserved bits, wrong lengths or stale frames must be reported as unavailable.

API	Period	Eight-byte payload
0x120 status	10 ms	version=4, sequence, debounced input mask, raw mask, channels=3, capabilities=0x7F, flags, reset flags
0x121 counts 1/2	100 ms	two uint32 rising-edge totals
0x122 count 3	100 ms	uint32 total, uint32 reserved=0
0x130 diagnostics	100 ms	uint32 transmit drops, TEC byte, REC byte, LEC byte, CAN flags byte
0x132 input state	Same as status	version=4, valid mask, floating mask, classified HIGH mask, uint32 input configuration
0x131 identity	1000 ms	4, firmware major=0, minor=9, patch=0, hardware=8, device ID, debounce ms, inversion mask

Status flags: configured=1, clock healthy=2, CAN initialized=4, bus-off=8, qualified 12 V power-good=16, qualified 5 V no-fault=32, all input classifications settled=64. Reset flags: power-on=1, pin reset=2, watchdog=4, software=8. Capabilities bits 0-4 indicate edge counts, runtime configuration, UART persistence, standalone hardware power and broadcast disable. Bit 5 indicates rail-health reporting capability; bit 6 indicates software bias/classification. Capabilities are static; flags bits 4/5 carry live qualified indications. The raw mask is the classified HIGH mask, excluding float probe transitions; check the valid mask in 0x132 before interpreting it. The debounced mask applies the active-state condition, inversion and debounce. Sequence is an unsigned byte and wraps. Counters wrap at 2^32 and clear on reset; runtime configuration preserves totals.

Diagnostic LEC is the STM32 three-bit last-error code (0-7); CAN flags are bus-off=1 and error-passive=2. CAN service INFO 10 reports measured 12 V/5 V in millivolts; INFO 11 reports enabled/good/latched/pending rail masks. ADC calibration and physical accuracy need bench verification. Normal status reports qualified rail health; it is not current telemetry. Raw masks remain diagnostic data during faults.

A bad rail indication clears its flag at the next 1 ms sample; restoration needs 100 consecutive good samples. During faults and recovery, firmware re-baselines input debounce and freezes edge totals, preventing supply collapse/recovery from adding counts. All ports require both rail indications because sensor-power cap positions are not readable. Java/C++ input helpers also require fresh telemetry, configuration, clock health and CAN health and all classifications settled. They cannot detect every failed/disconnected sensor or open cable.

Command API	Payload	Effect
0x220 runtime input config	`53 43 04 invert debounce_lo debounce_hi 00 00`	invert 0-7; debounce 1-100 ms; volatile, does not change ID
0x222 runtime bias	`53 43 04 00` + uint32 configuration	Volatile, re-baselines input classification; read back 0x132
0x221 resume telemetry	`53 43 04 00 00 00 00 00`	Explicitly resume after broadcast disable

Extended arbitration ID **0** is the FRC broadcast disable. Firmware stops telemetry and aborts pending transmit mailboxes immediately on processing it. Sampling and standalone sensor power continue; there are no actuator outputs. Resume requires the explicit command above, UART `RESUME`, or reset. Drivers never resume from a read loop. Integrate explicit resume with a reviewed robot lifecycle policy; this implementation still requires robot-bus testing and does not establish competition approval. See WPILib CAN addressing.

Read timeouts: status 100 ms, counters/diagnostics 350 ms, identity 1500 ms. Counters 1/2 and 3 are separate snapshots. Sampling is 1 kHz; debounce requires a continuously stable logical state for the selected interval. This is not a high-speed pulse counter or encoder.

UART: 115200 baud, 8N1, 3.3 V logic. Commands `INFO`, `HELP`, `SET id invert debounce_ms`, `RESUME`. `SET` saves a version-4 redundant CRC-protected record and resets. The old four-argument command and earlier flash records are rejected; provision Rev H explicitly. No power-control command exists.

Service extension in firmware 0.9.0

Rev H retains protocol 4 with hardware identity 8; use only the matching drivers. APIs 0x300/0x301 at address 0 are reserved for CAN service. Opening maintenance pauses telemetry/counting on participating SC12 nodes; treat samples as unavailable immediately in robot code. Status period can be 10/20/50 ms until reset.

Per-input configuration word

Bits 0-2, 3-5 and 6-8 select channels 1-3 mode: 0=OFF, 1=PULL_HIGH, 2=PULL_LOW, 3=FLOAT_DETECT, 4=BUS_KEEPER. Bits 9-11, 12-14 and 15-17 select each active condition: 0=HIGH, 1=LOW, 2=DRIVEN, 3=FLOATING, 4=NOT_HIGH, 5=NOT_LOW. Bits 18-31 must be zero; other enum values are invalid. Default word zero means push-pull/HIGH on all channels.

Input state 0x132: a cleared valid bit means UNKNOWN regardless of the retained high/floating bit. For a valid channel, floating takes its own state, high means HIGH, otherwise LOW. High/floating masks never overlap. Validity/classification precedes active-state mapping, inversion and debounce. See mode timing and electrical limits.

At the default 10 ms period, telemetry is 231 frames/s; 20/50 ms give 131/71 frames/s. Status and input state are separate snapshots. Read timeouts are 100 ms for both; 350 ms for counts/diagnostics and 1500 ms for identity. Runtime configuration does not reset counters or persist to flash. Measurements settle after a mode change before further edges are counted.

10 CAN setup and updates

Source: docs/CAN_SERVICE.md

Rev H firmware **0.9.0**, bootloader **1.1.0**, service protocol **1**. Normal sensor messages remain protocol **4**, hardware **8**. Implemented and software-tested; actual STM32/robot-controller, power-cut and vibration qualification is pending. This is our own service protocol, not GrappleHook-compatible firmware.

First installation and recovery

Install factory.hex once through SWD using program_factory.tcl. This combines the protected bootloader, relocated application and image manifest. Rev H requires its own factory image and fresh configuration. D/D2 hardware cannot gain the new bias circuit through firmware. The script preserves the two configuration pages and refuses an unexpected device/flash size or an already protected bootloader. Its hardware execution is a pilot qualification gate.

From `current`, with J8 wired per the custom mapping and a separately powered, current-limited board:

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -f tools/program_factory.tcl
```

Power cycle afterward and verify `boot_protected: true` using the tool. Write protection covers flash pages 0-7 (OpenOCD protection blocks 0-1). Updating requires those blocks locked and application/configuration blocks writable. The script does not change readout protection or mass-erase configuration. Do not program `sensor_can.elf` or `.hex` alone as an initial install: these are relocated application-only files without the boot-valid manifest. Do not upload `factory.hex` over CAN.

Normal CAN updates use application.sc12. If interrupted, retry the complete file: there is no old-image rollback or cross-session resume. Invalid images remain in the bootloader. A normal boot gives a 500 ms discovery opportunity; an explicit boot request stays there, and three successive watchdog resets before two healthy application seconds also request recovery. `run` exits a valid bootloader image. SWD remains the recovery path for bootloader/option-byte damage or a nonfunctional CAN connection.

Laptop and robot use

Use Python 3.10+; the robot tunnel transport needs only the standard library. SocketCAN additionally needs `python-can`; configure the adapter for classical CAN at 1 Mbit/s. Tk is required only for the desktop window (`python3-tk` on many Linux distributions).

The robot bridge runs in a background thread alongside robot code. Copy `ServiceBridge.java` and the adapter for your WPILib year from examples/bridge. It listens only on `127.0.0.1:5810`; access it over an authenticated SSH tunnel. The 2026 adapter requires an explicit maintenance-request flag **and** Driver Station disabled state for every transmission. The 2027 alpha-7 adapter uses `RobotState` and an explicit `CANPort`, compiled separately with JDK 25. Choose a classical-CAN-compatible bus; this MCU cannot transmit CAN FD.

Keep the robot disabled, mechanically secure it and keep outputs inhibited in your robot code throughout maintenance. The bridge rejects or disconnects when permission/disabled state is lost; it cannot prevent someone enabling the robot at the Driver Station. No robot program is stopped or redeployed by this tool. Do not expose this unauthenticated TCP framing directly on a network. The sensor protocol itself has no authentication; CRC detects corruption, not a malicious uploader.

Example using a roboRIO reached over USB; use your controller's actual address/account:

```
ssh -N -L 5810:127.0.0.1:5810 lvuser@172.22.11.2
python host/can_service.py --bridge localhost:5810 discover
python host/service_gui.py
```

Copy the desired 24-hex-digit UID from discovery. Replace `UID` below with it:

```
python host/can_service.py --bridge localhost:5810 blink --uid UID
python host/can_service.py --bridge localhost:5810 configure \
  --uid UID --device 17 --invert 0 --debounce 2 --modes npn,pnp,push-pull
python host/can_service.py --bridge localhost:5810 update \
  --uid UID --file firmware/build/application.sc12
```

Direct SocketCAN uses `--channel can0`. Other commands: `info`, `run`, `rate --milliseconds 10|20|50`. Settings persist after save/reset; status period resets to 10 ms. Existing driver timeout is 100 ms. Traffic is 231/131/71

frames/s at 10/20/50 ms; verify bus jitter physically.

Discovery resolves the complete STM32 96-bit UID independently of assigned CAN IDs, including blank and duplicate-ID devices. Blink confirms physical selection. Before a change, the tool reads back all three UID words. ID conflict checks cover **responding SC12 devices only**; a powered-off board or another Team Use product can still have a conflicting address. Use one maintenance client per CAN network. The service APIs must be reserved against other Team Use devices.

Opening a session suppresses normal telemetry on all participating SC12 boards, and freezes/rebaselines input counts. Application rail supervision continues during maintenance; the bootloader keeps both rails disabled. Drivers may retain their last sample until its normal freshness timeout; robot code must treat maintenance as unavailable immediately. Close/lease expiry restores normal sampling/telemetry unless Broadcast Disable previously latched it off. Broadcast Disable cancels maintenance and pending transmissions; explicit resume remains necessary for that latch. Never interpret a missing sample as a false input.

Wire protocol

Classical CAN, extended identifiers, eight data bytes, no RTR. Request ID `0x0A08C000`, response ID `0x0A08C040` (Team Use 8, miscellaneous type 10, APIs 0x300/0x301, address 0). Payload: opcode byte, sequence byte, nonzero little-endian session token (16 bits), little-endian value (32 bits). Response opcode OR 0x80 means success; OR 0xC0 means error. Echo sequence/token and validate the expected value. Errors 1-5 mean argument, state/offset, flash, image, and missing boot protection. No reply means an unmatched session/selection or transport failure.

Operation	Value / result
OPEN 0x01	Magic 0x314D4353; broadcast ACKs are identical; five-second lease
PREFIX0/1/2 0x02-04	Three raw UID words; no ACK; each change clears selection
PROBE 0x05	Prefix length 0-96, MSB first within each UID byte; matching nodes reply 1; only 96 selects
CLOSE 0x06	End session, no ACK
INFO 0x10	Index 0 flags/versions, 1 settings, 2 mode version, 3 upload offset, 4 capacity, 5 image validity, 6-8 UID words, 9 persistent input configuration, 10 rail millivolts, 11 rail states
BLINK 0x11	Zero; three-second identify LED
SAVE 0x12	ID/invert/debounce/reserved bytes; reserved must be zero; explicit save then reset
BOOT / RUN 0x13/14	Zero; enter bootloader / run verified application
RATE 0x15	10, 20 or 50 ms; volatile status period
INPUTS 0x16	Stage validated per-input configuration; commit with SAVE; cleared on new selection/session
POWER 0x17	Application only, selected UID: mask 1=12 V, 2=5 V, 3=both; disable then restart requested rails
SIZE / CRC 0x20/21	Padded application length and CRC32; bootloader only
BEGIN 0x22	Product/layout tag 0x53433138; invalidate/read back manifest before application writes
OFFSET / DATA 0x23/24	Byte offset then four image bytes; ACK next offset after flash readback
FINISH 0x25	Zero; validate complete flash CRC/vectors, commit manifest last, ACK image CRC

INFO index 0 packs service version in bits 0-7, bootloader mode in bit 8, configured in bit 9, boot protection in bit 10, application minor 9 in bits 16-23, and hardware 8 in bits 24-31. Index 1 packs ID/invert/debounce/status-period as bytes; unconfigured ID is 255. Index 2 is major/minor/patch in bits 16/8/0. Index 5 returns 0 or 1.

INFO index 10 packs measured 12 V millivolts in bits 0-15 and 5 V millivolts in bits 16-31. Index 11 packs enabled/good/latched/pending masks in bytes 0/1/2/3; mask bit 0 is 12 V and bit 1 is 5 V. Bootloader replies do not imply powered rails. Millivolt data is ADC-based and needs physical calibration verification.

After removing a sensor-power fault, restart only the required rail:

```
python host/can_service.py --bridge localhost:5810 power-reset --uid UID --rail 12v
```

`--rail 5v` and `--rail both` are also supported. The desktop tool exposes **Reset power fault** and displays both voltages. Reset is explicit; the application does not repeatedly retry a short. Firmware uses 10/60 ms off-to-enable delays and requalifies the rail before reporting usable inputs. A persistent fault latches off again. The outputs have no active discharge, so this clears protection without guaranteeing a complete power cycle of every sensor.

Host discovery walks binary UID prefixes; matching responders send identical presence frames. Prefix frames are paced; UID readback guards selection before writes. Queries retry four times, probes twice; default pacing is at least 6 ms between requests and a 300 ms ACK timeout. Binary discovery can take seconds per device; intermittent connectivity can omit a node, so repeat discovery before commissioning. Uploads send OFFSET and DATA for each four-byte word, bounding throughput and working around flash stalls. Only the immediately previous exact request is cached; offset/data logic also accepts an identical last word without rewriting it. Gaps, changed duplicate words and stale sessions are rejected.

Image and memory layout

Region	Address / size
Write-protected bootloader	0x08000000 / 8 KiB
Application	0x08002000 / 21 KiB
Image manifest	0x08007400 / 1 KiB
Configuration A/B	0x08007800 / 0x08007C00, 1 KiB each
Reserved SRAM	First 256 bytes: 192-byte vector table, validated recovery cookie

Both linkers reserve at least 1 KiB stack, excluding static data; runtime stack high-water measurement remains pending. The Cortex-M0 application copies vectors to SRAM and remaps them; bootloader handoff clears pending interrupts, stops SysTick/CAN and returns clocks to HSI. Compiled-register tests check vector copying, GPIO/CAN and configuration writes against peripheral models; they do not emulate analog effects or real interrupt timing.

The update file uses `<8s6I`: `SC12UPD1`, format 1, hardware 8, application address, padded length, CRC32, packed version, then image bytes. Length is 192-21504 bytes and a multiple of four. Initial SP must be aligned and within application SRAM, and Thumb reset PC inside application executable bounds. Manifest words are magic 0x49324353, size, CRC, product tag, CRC of the first four words, and final commit 0xC035A19E. Boot verifies manifest, vectors and full application CRC. No update command can write bootloader, configuration or option-byte regions.

Evidence and remaining gates

Tests execute the actual portable C engine with simulated flash, including 2,148 halfword erase/program power-cut positions on a two-page image, retries, corrupt files, wrong sizes, gaps, stale sessions, duplicate IDs and running-version verification. Compiled ARM tests exercise register intent and 34 configuration halfword interruption cases. Real TCP tests cover fragmentation and permission revocation with mock CAN. The Python uploader/GUI, bridge and firmware need end-to-end testing on real hardware before robot use. See qualification and unit programming/test fixture.

The `--modes` option and GUI persist independent input modes and active conditions using INFO 9 / INPUTS 0x16. Omitted modes preserve the existing persistent word; a runtime 0x222 change is not implicitly saved. SAVE commits ID, inversion, debounce and staged input settings together. See input modes.

11 Robot integration

Source: examples/README.md

Java 17 and C++ examples target WPILib 2026.2.1. Rev H uses protocol **4**, hardware **8** and firmware **0.9.0**. Use the matching `SensorCan12` and `SensorCan12Codec` files; older revisions are rejected. Provision a unique device ID through CAN service or UART.

Choose each sensor's 5 V / 12 V supply using the physical jumper before operation; the drivers do not select port voltage. See the Rev H setup guide and wiring.

CAN discovery, persistent settings and firmware updates are provided by the new service tools. The robot bridge has separately compiled 2026 and 2027 alpha-7 adapters. Sensor driver classes below remain pinned to 2026.2.1; the 2027 bridge does not port the normal drivers. Physical end-to-end CAN testing is pending.

```
private final SensorCan12 sensorBoard = new SensorCan12(17);
// In periodic(): absent means unavailable or a reported fault.
var detected = sensorBoard.input(1);
if (detected.isPresent()) {
  boolean sensorActive = detected.get();
  // Use a reviewed subsystem policy for this observation.
}
```

C++: `sensorcan12::SensorCan12 board{17};` then `board.ReadInput(1)` returns `std::optional<bool>`. Construct the driver once; an empty result must not silently become a valid “no object” observation. The helper requires fresh status, configured identity state, healthy clock/CAN and both qualified power indications. Raw `status()` / `ReadStatus()` remains available to diagnose faults. Counts are frozen during power faults and wrap at 2^32; compare/reset policy in the robot application.

Reads do not transmit or resume telemetry. `configureRuntime` / `ConfigureRuntime` sends volatile input inversion/debounce; it does not set persistent ID. `resumeTelemetry` / `ResumeTelemetry` is an explicit action after investigating broadcast disable, not a periodic automatic retry. Status timeout is 100 ms; count/diagnostic timeout 350 ms; identity timeout 1500 ms.

The Python host tool supports UART `info`, `configure`, `resume` and passive CAN `monitor`. UART uses 3.3 V levels at 115200 8N1; disconnect adapter power. CAN monitor requires a suitable external CAN interface; a UART adapter is not a CAN interface. Run the following from `current/`:

```
source tools/env.sh
python host/sensor_can.py configure --port /dev/ttyUSB0 --device 17 --invert 1 --debounce 2
python host/sensor_can.py monitor --interface socketcan --channel can0 --device 17
bash scripts/test_drivers.sh
```

Tests cover codec boundaries, all reserved status bits, stale/malformed responses, fault rejection, unsigned counts and explicit-only writes. The C++ example compiles against the actual pinned WPILib headers; Java compiles against pinned WPILib jars. Hardware bus behavior and competition acceptance remain open qualification steps. See protocol, wiring, and validation.

Independent input modes

Use the desktop service tool to persist per-input settings. Runtime Java example (volatile; verify `inputState()` readback):

```
sensor.configureInputsRuntime(
    new SensorCan12Codec.InputMode[] {
        SensorCan12Codec.InputMode.PULL_HIGH,
        SensorCan12Codec.InputMode.PULL_LOW,
        SensorCan12Codec.InputMode.OFF},
    new SensorCan12Codec.ActiveWhen[] {
        SensorCan12Codec.ActiveWhen.LOW,
        SensorCan12Codec.ActiveWhen.HIGH,
        SensorCan12Codec.ActiveWhen.HIGH});
```

C++ exposes `ConfigureInputsRuntime` with `std::array<InputMode,3>` and `std::array<ActiveWhen,3>`, and `ReadInputState()` for classification/configuration. `input()` / `ReadInput()` require fresh, healthy status and all inputs settled. Level helpers return UNKNOWN when the channel valid bit is clear. See mode behavior and electrical limits.

12 Purchasing and factory quote

Source: docs/FACTORY_ORDER.md

Rev H native CAD, BOM, firmware and individual manufacturing exports include the RGB/header change. A refreshed full-validation factory ZIP has not been issued; see package status. The extra power rerun was canceled before completion, while the focused LED/header checks passed. Previous Rev G ZIPs are historical and do not contain the new LED/header.

For quotation, use BOM_JLCPCB.csv, CPL_JLCPCB.csv, the fabrication exports and Factory_BOM.csv for exact-MPN matching. Read the engineering review and physical acceptance procedure. No factory quotation or order is included.

Fabrication

Setting	Requirement
Board	52 × 74 mm, four copper layers, FR-4, 1.6 mm nominal
Layer order	F.Cu signal/power, In1.Cu GND, In2.Cu GND with short/slow control and service routes, B.Cu signal/power
Copper	35 µm nominal each layer; verify finished copper with supplier
Dielectric	Symmetric four-layer stack; target outer-to-inner dielectric ≤0.25 mm; supplier must confirm final stack
Finish	ENIG retained for fine-pitch IC assembly
Minimum trace	0.18 mm
Clearance	0.20 mm general; U8 package pad-to-pad 0.15 mm local minimum
Vias	Through vias, minimum 0.30 mm drill / 0.60 mm pad; no blind/buried vias
Drills	Separate PTH and NPTH Excellon; mounting holes remain nonplated
Solder mask	Both sides; preserve fine-pitch openings; review U8 mask webs with CAM
Stencil	Top only, nominal 0.10 mm laser-cut; use supplied paste apertures
Panels	Factory rails/fiducials may be added outside the delivered outline; preserve connector overhang

Native nominal stack: 0.035 mm copper on all four layers, 0.20 / 1.04 / 0.20 mm dielectric and 0.01 mm solder mask per side, totaling 1.60 mm. These are planning thicknesses; final fabrication tolerances and material selection belong to the approved supplier stack.

This stack is an order requirement, not a confirmed laminate/impedance quotation. CAN is short board-level routing; no fabricated impedance coupon has been qualified. Get the supplier's CAM/stackup review before fabrication.

Assembly

Factory scope: 130 SMT components on the top side only. Use `bom/Factory_BOM.csv` (exact MPNs) and `output/assembly/CPL_JLCPCB.csv`. The simplified `bom/BOM_JLCPCB.csv` is a matching aid; blank LCSC numbers require exact-MPN matching against Factory_BOM, never an automatic substitution.

User scope: six terminal blocks, three voltage headers and one J8 service header (ten through-hole parts), then three loose caps. The factory must leave J1-J6, J8 and JP101/JP201/JP301 empty. `positions_all_top.csv`, the complete BOM and drawings describe the finished board, not additional factory placements.

CPL coordinates are mm from the PCB lower-left, positive X right and Y up, with native KiCad counterclockwise rotation. Match the uploaded preview to the assembly drawing and check IC/diode pin 1; machine-library

rotation conventions can differ.

Power headers are unpopulated; the JP2 termination bridge and all J8 holes are **unbridged/unpasted as supplied**. The factory must not add solder across JP2 unless a separate reviewed port-configuration work instruction is supplied. The user will hand solder all through-hole parts after delivery. The family name “LSF-SMT” does not make these exact connectors ordinary two-pad SMT parts.

Use exact TPS61085DGKR for U8, latch-off TPS2553DBVR-1 for U4 and TPS259620DDAR for U9. U9 has an exposed ground pad with supplied segmented paste apertures and six adjacent thermal vias. Inspect its solder attachment. Do not substitute auto-retry TPS2553, a lower-saturation inductor, or nominally equal small MLCCs without electrical review. Preserve capacitor voltage/dielectric/DC-bias behavior, the precision resistors and the 13 kΩ / 33 nF compensation network.

D10 is OPSCO SK6805-EC15 (C2890035), with the exact supplied A/1 pin layout; U10 is SN74AHCT1G125DBVR. Inspect VSSOP/MCU lead joints, diode polarity and all passive values. Perform unpowered shorts/selector checks, staged current-limited bring-up, firmware programming/readback and rail/input/CAN tests. Track every component substitution and lot.

Sourcing, cost and delivery

Cost reduction and batch estimates separates parts procurement from assembly. LCSC inventory is not proof of JLCPCB factory allocation. No SMT part allocation, complete cart or final quotation has been confirmed. Request exact-MPN matching and incoming date confirmation before paying; the STM32 and any unavailable exact MPN may require outside procurement. Any unapproved item must remain unresolved rather than being silently substituted.

Request Economic PCBA first if the exact parts and four-layer board qualify; compare Standard PCBA with the same SMT-only scope. Avoid paying for bottom-side assembly or through-hole work. Standard assembly still charges feeder loading for all 55 SMT types. No programming or functional-test service is included by this package unless separately quoted; the user performs bring-up and programming.

One-month delivery is unconfirmed. Obtain allocated-part dates, CAM/assembly timing and a destination-specific tracked express quote before scheduling the robot build. Procurement holds and customs may exceed the target. Order hand-fitted parts in parallel once the BOM is finalized.

Programming deliverables

The packet includes combined `firmware/build/factory.hex` and `tools/program_factory.tcl` for initial SWD programming/protection. Power cycle and verify protection, then use CAN for setup and later `application.sc12` updates. User programming is the default quoted scope; see fixture procedure. Do not substitute an application-only ELF/HEX.

13 Qualification

Source: docs/VALIDATION.md

No Rev H prototype has been assembled, powered or tested. The automated evidence checks the design, software and manufacturing package. Physical acceptance remains a separate step.

Automated evidence

Native KiCad ERC/DRC, explicit model/schematic/PCB parity, assembly constraints and independent reconstruction of all four Gerber copper layers check the delivered geometry. Firmware tests exercise the application state machine, CAN servicing, ADC scaling and register-level GPIO behavior; host and Java/C++ tests check protocol handling. Mocked hardware does not prove physical STM32 or CAN behavior.

Electrical calculations cover input thresholds, sensor leakage/residual-voltage margins, divider stress, capacitor derating, converter current budgets and output current-limit tolerances. The rail-reading budget reserves ±2.5% gain and ±90 mV offset at the sensor rail; firmware health bands include this uncertainty. The averaged boost loop screen covers 768 CCM corners. See the release report for the exact retained results and hashes.

Manufacturer-model transient screen

Current run status: eight standalone/load-step cases passed or reused matching completed traces. The two coupled startup runs were canceled before 40 ms, so the H full-suite result is incomplete. The following describes the test design and historical G findings, not a completed H startup validation. The complete G evidence remains in the archive.

The harness maps the fitted TPS61085 and TPS259620 pins/components into original TI models, with the Diodes B240A rectifier model. Results retain every scenario, criterion and generated netlist; archived Rev G waveforms show the responses. A failed criterion returns nonzero and prevents the release validation command from passing.

Scenario	Engineering question
Low-input combined startup, 66 µF raw bank, 100 µF external and 300 mA	Does controlled load connection avoid the former input-limiter startup failure?
30→310→410→0 mA boost load	Does the converter regulate at normal load and the output limiter's upper current bound?
Running short and explicit reset	Does output protection limit current and recover when commanded?
Startup into a short and explicit reset	Does the circuit remain bounded without requiring a successful first startup?
470 µF external capacitance with 300 mA	Does a larger capacitive load start in the standalone eFuse model?
5 V startup, short and enable reset with concurrent 1.3 A load on internal 5 V	Does the unchanged TPS2553 latch-off branch work with its new GPIO enable?
Half-timestep repeat of the running 12 V short	Are the fault/reset results numerically stable away from imposed switching edges?
Sustained short, removal, then enable cycle	Does the thermal-latch model remain off until deliberate recovery?
Reduced current-limit surrogate, 120 µF external and 300 mA	Does the supported capacitance plus 20% start with minimum charging-current headroom?

Capacitance stress cases use constant nameplate values; ceramic DC-bias reduction makes actual capacitance voltage-dependent. This is a conservative capacitive-load screen, not a guaranteed sensor-capacitance rating. Actual sensors can have constant-power or nonlinear startup loads.

The reduced-limit case uses a 2.92 kΩ test resistor in the typical eFuse model, producing approximately 314.4 mA, below the calculated 314.7 mA minimum for the fitted 2.61 kΩ resistor. This deliberately harder charging condition reaches 11.4 V in about 93 ms with no model fault. It supplements the analytical startup budget; it does not represent a full silicon or temperature corner model, and does not change the BOM.

The upstream supply is idealized with 120 mΩ series resistance, 25.92 µF internal capacitance and 450 mA concurrent logic/RGB/5 V load. The encrypted LMR51420 vendor model cannot be used in this ngspice flow. Its control loop/current limit, PCB/cable inductance, robot brownouts, temperature and production corners are not simulated. The default-off transistor stage is represented by an idealized enable switch; its physical startup behavior needs the separate bench check below. The eFuse thermal timing comes from its behavioral model, not a board thermal solution.

The normal boost case uses fixed derated C60/reservoir values. Silicon models are typical 27°C; inductor DCR is 32 mΩ and inductance is linear below the specified 8 A/20%-drop domain. D5 uses the vendor diode model; D6 uses an approximate static TVS clamp. This does not establish fast fault overshoot or reverse-power survival.

PSpice adaptations are syntax-only: namespacing, duplicate PARAMS removal and Boolean XOR translation. The combined startup uses KLU/Gear with `xtrtol=2`, a 50 ns maximum step and a 1 fF numerical node shunt. A complete repeat uses `xtrtol=7` and a 25 ns maximum step. Both retain relative tolerance 0.003 and the same circuit/models, and must pass all electrical criteria. Over the full 40 ms, the comparison requires voltage differences below 50 mV and peak-inductor-current difference below 20 mA. These are solver controls, not component changes.

The original `xtrtol=1` combined run stopped numerically at 28.971 ms during the transition into regulation. It and other incomplete solver diagnostics are excluded from passing evidence; an incomplete trace cannot establish circuit operation or hardware failure. Standalone enable conductance ramps over 1 µs; a discontinuous ideal enable switch also produced a numerical timestep failure. The finite transition keeps the same 2 Ω final shunt. Zero-supply operating-point warnings are accepted only after successful transient-OP fallback and full completion. No semiconductor parameter is tuned to obtain a pass. Results have not been correlated with PSpice or a physical board. Rev F's superseded failure reports remain in its immutable archive.

Prototype gates before production release

Gate	Measurement / acceptance intent
Assembly	Confirm exact horizontal terminals, pin-to-hole fit, polarity, U8 VSSOP joints, D4 polarity, U9 TPS259620DDAR and exposed-pad soldering, selector isolation and no solder shorts
Staged power	No sensors first; current-limited 12 V source, verify 5 V/3.3 V before full operation, then regulated 12 V and low-load input current
Regulation	Sweep 6-22 V with 0-300 mA on each rail separately and together; verify 12 V remains within the selected sensors' supply limits at their cable ends
Cascaded startup	Scope V5_INTERNAL, BOOST_12V, BOOST_SS, SENSOR_12V and 3V3 during slow/fast input ramps, repeated starts and brownouts. Confirm boost startup/inrush does not falsely latch U9 or collapse U1; review operation below 6 V
Transients	6↔14↔22 V line steps; 0↔300 mA load steps; representative sensor startup/inrush and cable lengths; check ringing/undershoot on both cascaded converters
Stability	Measure loop response or justified equivalent transient qualification on the cascaded 5 V buck / 12 V boost, including CCM and light-load/DCM modes; target >45° phase and >10 dB gain margin where measured
Faults	Short each sensor supply and each port; verify current-limit behavior, hardware current limiting, firmware latching and explicit CAN reset, firmware flags, no spurious counts and safe temperatures. Measure peak current and output-capacitor discharge energy before U9 disconnects; verify cold-start into a short and hot-plug loads
Electrical protection	Reverse input, realistic protected-branch fault energy, fuse clearing/I^2t, inductive transients and TVS stress using suitable test equipment. Not automotive load-dump qualified
Temperature	Both rails fully loaded at 6/12/22 V, intended ambient and later enclosure; measure U8/U1/L1/L2/Q1/fuse/terminal and MCU temperatures, hot-start/restart and surface safety
Inputs	0-30 V PNP/NPN/push-pull thresholds, off-state leakage/residual voltage, ground offset, sensor disconnection, wiring transients and all intended sensor models
CAN/firmware	Real roboRIO bus at 1 Mbit/s with correct end termination; no normal telemetry before provisioning, unique IDs, bus-off, clock faults, watchdog, broadcast disable, explicit resume, stale data and power-fault handling
EMC/mechanical	Motor-noise environment, ESD at terminals, cable retention, vibration, mounting insulation and programming-adapter fit; repeat after enclosure design
Production process	Approved stackup/CAM, stencil/profile, supplier alternates, test fixture, serial/lot records, programming/provisioning and final functional limits

FRC integration

Use a single appropriate protected power-distribution branch and wiring sized for its protection. The 0451 3 A fuse has a published 300 A interrupt rating at 32 V DC; confirm prospective branch fault current and clearing energy for the actual harness. The onboard fuse is secondary protection, not permission to bypass the robot distribution rules. Sensor grounds are common with the board; isolate conductive sensor cases where required by frame-ground rules. Do not use dedicated radio/roboRIO power outputs as a convenient supply.

The custom Team Use CAN address uses manufacturer 8 and miscellaneous device type 10. The implementation processes broadcast disable ID 0; it has no actuator outputs. It is not a CTRE Phoenix device. Robot inspection and the current season's rules still apply. Commercial CAN product acceptance, assigned identifiers and required ecosystem support are separate outstanding steps; Java/C++ examples alone do not establish approval.

Primary references: 2026 FIRST manual, especially custom-circuit voltage, distribution/wire and frame-ground rules; WPILib CAN addressing and acceptance requirements. Recheck applicable rules for the actual event/season.

Rev H-specific prototype checks

Measure the 5 V/12 V ADC readings against a calibrated meter throughout the supply and load sweep. Check 11:1 divider scaling, VDDA compensation, settling and fault thresholds. Confirm both rails stay disabled during SWD programming, bootloader operation and MCU reset. Remove 3.3 V while raw boost remains energized and verify the Q2/Q3 stage holds the 12 V supply switch off. There is no active output discharge: distinguish residual capacitor charge from a conducting switch, using a known test load and probing EN. Check the six U9 ground thermal vias and exposed-pad solder joint.

Confirm the output eFuse nominal 357 mA limit, tolerance bounds, load-capacitance startup and thermal latch behavior. Verify a short on either branch does not cause unsafe shared-supply oscillation. The 768-corner CCM screen uses assumed amplifier/modulator bounds and does not qualify silicon behavior.

Verify the exact 2 mm header and cap fit, both 5 V / 12 V positions and unpowered open state on every channel. Verify no connection between the two rails in either position. Check solder fill on all 49 user-fitted through-hole joints. Fit/retention, vibration and impact testing must include the actual caps, chosen removable retention compound, enclosure and robot harness; these headers have no positive latch. Compare the substituted passive values/ratings against the supplied BOM. Factory BOM and CPL must contain exactly 130 SMT references, with all ten through-hole components omitted. Firmware identity is hardware 8 / protocol 4; prior-revision firmware is rejected.

CAN servicing qualification

The service engine and host uploader pass simulated flash cuts/retries; ARM register checks and both bridge adapters compile. Before robot use, measure actual boot/vector/interrupt behavior, option-byte protection, stack use, upload time, CAN load, bus-off recovery and duplicate-ID discovery. Interrupt power during invalidation, every erase/program phase and final commit on physical units. Verify persistent configuration, bootloader recovery, controller disconnect/enable rejection and healthy telemetry after exit. Protocol and evidence; per-unit fixture records. No software simulation closes these physical gates.

Electronic bias qualification

Verify Q101/201/301 PMOS and Q102/202/302 NMOS polarity and correct exact MPNs. Scope all six gate signals during reset, SWD programming, bootloader, watchdog recovery and supply ramps: pull-up gates HIGH and pull-down gates LOW mean both biases off. Verify mixed NPN/PNP/push-pull operation and nonvolatile readback independently on all three channels.

Sweep external signal 0-30 V in every mode. Measure inactive sensor leakage margins, active residual voltage and pull-up sink current (~3.5 mA maximum). Test 3.3 V and 5 V push-pull, unpowered sensors and unpowered board backfeed. Qualify the BAT46W leakage/rail-bleed budget over the actual operating temperature; current calculation uses its 60°C reverse-leakage specification, not a zero-backfeed claim.

Test floating detection with open contacts, NPN and PNP inactive outputs, long cables and added capacitance. Verify UNKNOWN during inconsistent probe pairs, no probe-generated edge counts, and one debounced edge on actual active-state transitions. Check keeper retention while released and LOW initialization after reset. Verify fixed input modes for faster signals; float detection can alias changing sources.

Rev H design envelope: at most 100 µF total external capacitance on the shared 12 V rail, and 22 µF on the shared 5 V rail, at 300 mA per rail. The startup calculation includes +20% capacitance and the current-limit minimum; the 470 µF simulation is a typical diagnostic beyond that envelope. Qualify nonlinear sensor inrush separately.

Rev H RGB and service header

The circuit regression compares H against G and permits only the status indicator, its buffer/passives and the physical service-header replacement. Power component values and sensor inputs are unchanged. Compiled ARM tests check TIM2 update DMA mapping, both 8/48 MHz pulse settings, GRB encoding, reset gaps, bounded transfer-error recovery and CAN acknowledgment freshness. The application harness checks status during normal, unplugged-CAN, initialization, clock and power-fault cases. These tests emulate CPU/register boundaries, not the actual timer/DMA silicon or emitted optical waveform.

Follow the fixture checks and status/programming guide on the first assembled unit. Verify LED package orientation and 5 V buffer margin, reset/brownout transitions, no sampling interruption, the five-wire programming harness, and the enclosure optical opening. The nominal STEP model is not physical-fit evidence.

14 Unit fixture

Source: docs/FIXTURE.md

Build a reusable fixture around the documented 2.54 mm J8 header and five mapped wires, ST-Link, a separately current-limited J1 supply, accessible rail test contacts, CAN adapter and three independently driven input fixtures. This document and scripts are ready for the pilot; no physical fixture or unit test has been performed.

Leave voltage caps absent for initial programming. Keep the SWD probe's voltage pin as target reference only. Program/verify the combined image and set boot write protection with program_factory.tcl, then power cycle and verify protection over CAN. Save the programmer log, factory image SHA-256, component lot and full MCU UID. The setup checks STM32F04x device ID 0x445, 32 KiB flash and WRP pages 0-7 (OpenOCD protection blocks 0-1, not 0-7), consistent with the OpenOCD STM32 driver. Qualify the exact probe/OpenOCD version on the first unit.

Configure a unique ID through the service tool. Record internal 5 V, logic 3.3 V, protected sensor 5 V and regulated 12 V with a calibrated meter; acceptance windows are 4.8-5.3, 3.15-3.45, 4.65-5.45 and 11.4-12.6 V. These generous unit-screening windows do not replace rail regulation/load/temperature qualification. Power off before fitting selector caps. Exercise each input low and high using the approved bias and compatible test signal; observe fresh protocol-v4 status and count transitions. Verify telemetry after closing maintenance. Power cycle again and verify saved ID/inversion/debounce and all three input modes/active conditions.

Copy fixture_measurements.template.json, enter measured voltages and set each check true only after actually observing it. Record live identity/protection plus those results:

```
python host/fixture_record.py --channel can0 --uid UID \
  --measurements measured-unit.json --operator NAME --lot LOT \
  --output unit-UID.json
```

Use `--bridge localhost:5810` for a robot tunnel. An incomplete/failed record exits nonzero. Existing output files are never overwritten. The JSON stores operator/lot/time/UID/settings and a hash of the supplied measurements; keep it with the programming log. It does not independently measure voltages or validate operator assertions.

Design qualification still needs 6/12/22 V input tests, both rails at load, shorts, temperatures, representative sensors/cables, vibration, actual flash power cuts, bus-off/controller disconnect and boot/interrupt timing. See VALIDATION.md. Complete the pilot before calling units production-ready or quoting a qualified manufacturing yield.

Exercise mixed software modes and verify floating detection without false counts using INPUT_MODES.md. Unsettled inputs must not appear as valid robot measurements.

Retained power qualification requires recorded checks for the U9 exposed-pad joint, rails off in bootloader/reset, ADC readback against a calibrated meter, and each rail shorted then recovered through addressed CAN reset. During first-article qualification also remove MCU power while raw boost is present; verify Q2/Q3 holds the 12 V output off. Use controlled test equipment and load fixtures for these checks; do not count an unperformed test as passed.

Rev H also requires checking all RGB colors against the legend, CAN unplug/recovery, boot/update/identify and both rail-fault indications. Measure DIN low/high levels, 800 kbit/s bit timing and reset-low interval >200 µs, including startup on 8 MHz fallback and boot-to-application handoff. Verify input sampling and CAN service continue during LED updates. Confirm header orientation and fit, remove the programming leads, and inspect LED visibility through the proposed case/light pipe. Record these observations with the unit log; no such physical test has yet been performed.

15 Connectors

Source: docs/CONNECTORS.md

References	Exact MPN	Series description	DigiKey
J1-J3	**1824420000**	LSF-SMT 3.50/02/90 3.5SN BK TU	281-2020-ND
J4-J6	**1824430000**	LSF-SMT 3.50/03/90 3.5SN BK TU	281-2021-ND

Manufacturer pages: 2-pole, 3-pole. `/90` gives horizontal wire entry parallel to the PCB. `/135` is the angled version and is not used. Phoenix PTSM is not fitted.

These are through-hole/THR parts despite “SMT” in the family name. Pitch is 3.5 mm, with two solder legs per pole and 7 mm row spacing. PCB drills are 1.1 mm. Housing widths are 7.7 mm / 11.2 mm. Manufacturer CAD is 14.3 mm deep; the footprint conservatively reserves 14.75 mm. Height is 8.5 mm above the mounting surface; nominal pin tips reach 3.5 mm below that surface (about 1.9 mm beyond a 1.6 mm PCB).

Current renders use the official STEP solids with a rigid translation (-2.13,-12.625,-3.5) mm into the footprint origin. They are not approximate connector boxes. Display colors are assigned by the project. Original files, transformed files, URLs, hashes, pin/drill checks and nominal bounds are in vendor_connectors.json. Script `import_connector_models.py` checks all six placed pad patterns and the two vendor models. Do not regenerate models merely to change routing: STEP exporters can alter file headers without changing geometry.

Top terminals face outward at the power/CAN edge; three sensor terminals face outward at the opposite edge. The board and nominal terminal bodies occupy a conservative 52 × 82.1 mm envelope; add room for cable bend, strain relief and button access. Renders do not validate conductor retention, tool clearance in an enclosure, molding tolerances or actual solder fit.

Rev H voltage headers

JP101/JP201/JP301 use Harwin M22-2010305: 2 mm pitch, 0.50 mm square pins, 6 × 2 mm body, nominal 5 mm height above the PCB and 3.4 mm tails. The project's 2 mm header footprint uses 0.85 mm drills in 1.35 mm pads, retaining a 0.25 mm nominal copper annulus, and a 0.25 mm nominal body courtyard margin. Pads 1-3 run left-to-right at Y=60.4 mm; pin-1 positions are X=5, 21 and 37 mm.

The Harwin drawing recommends 0.80 ±0.05 mm holes. The retained footprint adds 0.05 mm nominal diameter for hand assembly: with JLCPCB's normal −0.08/+0.13 mm hole tolerance, the minimum becomes 0.77 mm, above Harwin's 0.75 mm minimum. This trades a slightly larger solder gap for insertion clearance; verify finished holes and solder fill on the first article.

The Sullins SPN02SVEN-RC drawing specifies a closed top, 2 mm pitch, 0.50 mm square mating pins and nominal 4 × 2 × 4.92 mm body. Contact height is 2.45 mm from the entry, within the header’s 3.5 mm mating pin length. Nominal mating geometry agrees; physical fit and retention remain untested. Caps are delivered loose and not included in the assembled-board STEP model. Use the finished cap envelope for enclosure access/clearance and allow room to remove them.

The generic local header STEP represents the nominal pin/body geometry, not manufacturer-qualified tolerances. Vendor connector hashes continue to cover the unchanged six terminal models. See wiring for one-cap-per-header configuration and vibration retention qualification.

J8 is a hand-fitted XFCN PM254V-12-10P-H85 recessed female socket, 2×5 at 2.54 mm pitch, with 1.10 mm plated holes and 1.70 mm pads. The matching nominal model and the full custom pin map are in status and programming and wiring.

Edge clearance and visible markings

The bottom resistor R101 now lies horizontally between the first two sensor terminals: its closest pad is 2.20 mm from the board edge, previously 0.575 mm. R103 and R203 moved into the gap beside the third terminal, with 2.05 mm pad-to-edge clearance, previously 0.80 mm. F1, D102, R22 and R109 also moved inward. R104 has 1.98 mm clearance; the two small parts above J8 have 1.70 mm. The compact buck feedback pair R1/R2 retains its original 1.30 mm pad clearance and quiet local routing. These distances are to copper pads, not component bodies.

The board remains 52 × 74 mm, with all components on top. Keep an insulating enclosure around the board and underside solder tails. The recessed debug contacts reduce accidental bridging from above; the socket is unkeyed and is not sealed against metal debris. Remove the programming cable for robot use.

The power headers sit at Y=60.4 mm. Their 5V/12V labels share a baseline above the pins at Y=58.8 mm. The specified Sullins SPN02SVEN-RC cap has a nominal 4 × 2 mm footprint; both fitted positions leave the labels above it exposed. Front legends identify VIN, CAN L(G)/H(Y), termination, DEBUG, each 5V/12V selection, and V1/V2/V3, GND and IN1/IN2/IN3 at the sensor terminals. Both CAN pairs share a horizontal label baseline: G means green, Y means yellow; see wiring. The back repeats the full connection and custom debug map. Select voltages with power off; fit one cap per header.

The two bare pads beside TERM are JP2, the optional CAN termination solder bridge. Closing it enables the onboard 120 Ω resistor across CAN H/L; leave it open unless this board needs to terminate a bus end. The single round gold pad near VIN is TP1, connected to GND for a meter or test probe. Both features intentionally remain unpopulated.

The large cylindrical part near the left edge is C1, a Panasonic EEE-FK1H470P, 47 µF / 50 V, nominally 8 mm diameter and 6.2 mm body length. Its center is (5,21) mm, placing the nominal can edge 1 mm inside the PCB edge. The local CAD model is approximately 6.5 mm tall, below the 8.5 mm terminals. Engineering assessment: board orientation alone does not require moving C1, but its exposed side needs protection from impacts and cable forces. Keep clearance between the enclosure and capacitor; support the board and strain-relieve cables independently. Retain this placement for an enclosed prototype, pending physical vibration/impact checks. Panasonic's handling guidance requires evaluation in the actual vibration environment; this standard part must not be treated as a 30 G vibration-proof variant.

Layout refinement

The outline now has **1.5 mm corner radii**, within the same 52 × 74 mm dimensions. Mounting-hole centers, connector positions, service socket and the LED duct space stay fixed. All six input-bias transistors share the same Y=57.15 mm row; R4 aligns with the neighboring support parts. Native DRC and the independent Gerber check use the rounded contour, including copper-to-edge clearance around the corners.

LaserCAN's board informed the rounded outline and consistent grouping. Its approximately 25 × 25 mm board places components on both sides and uses a different connector and sensor arrangement. SC12 retains top-side assembly, hand-solderable terminals, three external input circuits and two protected sensor supplies. The refinement changes no purchased parts or functions; it does not claim a lower factory quote. No LaserCAN circuit or library files were copied into the design.

16 Cost reduction

Source: docs/COST_REDUCTION.md

Rev H adds the RGB indicator and inexpensive header programming to Rev G. Parts purchasing rises by approximately **$0.18 per board at ten boards**, from $28.81 to **$28.99**. The 12 V output eFuse, controlled startup, rail measurement and removable power jumpers remain unchanged. The resulting parts budget remains about **$6.15 below Rev E**.

Ten-board parts purchasing	Per board
Rev E	$35.14
Rev F	$27.17
Rev G	$28.81
Rev H	$28.99

The RGB LED and logic buffer cost about $0.20 together at the ten-board tier. Local bypass capacitors, a reset pulldown, series resistor and the hand-fitted J8 female socket complete the change. Removing the former green LED and sharing existing passive purchase packs offsets part of the added cost. See the exact circuit and procurement regression below; consumed-part prices and minimum-pack cash costs differ.

Boards	Parts purchasing per board	Lot parts purchasing
1	$55.14	$55.14
5	$33.18	$165.89
10	$28.99	$289.89
25	$26.57	$664.16

Supplier observations are dated September 8-9, 2026. Prices include minimum packs, six terminals, three voltage headers, one service header and three loose caps. PCB fabrication, assembly, attrition, programming, functional testing, freight, tax and duty are extra. At quantity one, consumed parts are $36.15; minimum purchase quantities raise the cash outlay to $55.14. These are estimates, not an allocated cart or factory quote.

Cost CSV · per-MPN purchases · supplier observations · revision comparison.

The female socket adds $0.044 per board at ten compared with the previous male header. Layout and silkscreen add no fitted parts, board area or second-side assembly.

Assembly and programming choices

There are **130 top-side SMT placements across 55 exact MPN types** and **ten hand-soldered through-hole parts (49 joints)**. Hand fit J1-J6, J8 and JP101/JP201/JP301. The output eFuse's exposed-pad paste pattern and thermal vias remain required.

Compared with G, the two additional SMT types add approximately $3.06 per Standard assembly order in feeder fees, or $0.31 per board at ten, before placement/inspection changes. If both are charged extended types under Economic assembly, the increment is approximately $0.61 per board. Exact eligibility and charges require a quote. The per-board $0.18 parts increment is therefore not a complete assembled-price increment. JLCPCB fee basis, checked 2026-09-09.

The recessed service socket avoids buying a dedicated pogo cable. Reuse one programmer and mapped harness for all boards; an existing suitable ST-Link can avoid buying a new probe. Programming options and one-time tool cost.

What double-sided assembly might change

The implemented board remains **52 × 74 mm**, with all components on top. Moving suitable small components underneath might reduce area by **15-25%**, approximately **52 × 56-63 mm**; a reasonable planning target is **52 × 60 mm, 19% smaller**. This is an engineering estimate, not a completed/routed alternative. Connectors, mounting clearances, hot power components and return-current paths limit the saving. The enclosure must also allow bottom component clearance.

At ten boards, budget **$3-6 extra per board** for assembly on both sides, assuming Standard assembly in both cases. JLCPCB's posted extra setup ($25.56) and extra stencil ($8.21) alone total $33.77 per order, or **$3.38 per board at ten**. Other fees, panelization and any change from Economic to Standard may change that substantially. Bare-PCB savings at this size may be small or zero within the same price bracket. Retain top-side assembly for this cost-sensitive ten-board batch unless the enclosure specifically needs the size reduction.

Quotation and delivery

Ask the Chinese assembler to quote top-side SMT only, with exact-MPN allocation and functional test priced separately. Compare Economic and Standard eligibility using the same scope. LCSC stock does not guarantee JLCPCB allocation; TPS259620 stock was limited when observed. No supplier has been contacted or order placed.

The assembly scenarios are partial fee calculations. One-month delivery requires allocated-part dates, an assembly slot and destination-specific freight/customs timing. No arrival date is confirmed. Factory instructions.

17 Tools

Source: TOOLS.md

Run the pre-fabrication review after refreshing the native board and electrical evidence: `python scripts/check_order_readiness.py`. Its `--verify-report` mode checks retained hashes and recalculates routed-copper voltage drop. It does not reroute the PCB or perform physical testing.

The saved native KiCad schematic/PCB are authoritative. Circuit data, schematic, BOM and PCB must agree. Historical placement/router experiments have been removed; validation never reroutes the board.

Tested toolchain: KiCad 10.0.6 and its matching pcbnew interpreter, Python 3.12, Arm GNU GCC 13.2.1, GNU make, host GCC/G++, Java 17, Java 25 for the WPILib 2027 adapter, Poppler, InteractiveHtmlBom 2.11.2 and pinned Python packages. Native tool binaries and dependency caches are outside Git. PDF generation uses DejaVu Sans/Mono.

```
# Run from current/, after installing native tools.
uv venv --python python3.12 .venv
uv pip sync --python .venv/bin/python tools-requirements.lock
source tools/env.sh
python tools/fetch_driver_dependencies.py
CANDI_JDK25=/path/to/jdk25 bash scripts/validate_release.sh
```

`tools/env.sh` uses this directory's venv, or the repository venv. It prepends `$CANDI_TOOLS_ROOT/bin` (default `$HOME/.local/share/candi-tools/bin`). Expected wrappers include `kicad-cli`, `kicad-python`, `host-gcc`, `host-g++`, `ibom` and Arm compiler commands. `kicad-python` must match the installed KiCad binary bindings. Java 17 can be selected with `CANDI_JDK`; the 2027 test downloads its pinned dependencies.

The validation script refreshes electrical budgets, loop sensitivity, board constraints, firmware/driver/bridge tests, native ERC/DRC/parity, manufacturing exports and independent Gerber geometry/connectivity checks. It compares against the small frozen Rev G contract in `hardware/design/rev_g_baseline.json`, without depending on an expanded historical checkout.

After validated changes, refresh documents and inspect the results:

```
bash scripts/render_board.sh
python scripts/compare_assembly_cost.py
python scripts/build_documents.py
python scripts/build_handbook.py
python scripts/review_documents.py
# Inspect final renders/PDFs and update reports/visual_review.json with their hashes.
python scripts/build_release.py
python scripts/verify_archive.py
```

`build_documents.py` creates the factory packet and Gerber ZIP after checking current release and power-review hashes. `build_release.py` also requires a completed visual-review record matching the final artifacts, then writes the source/review archive and manifest. `verify_archive.py` extracts it, verifies all hashes and reruns native checks. The manifest excludes itself and its enclosing ZIP from its own hashes. Review PDF page bounds and rendered pages; automated text bounds cannot detect every overlap.

Optional desktop setup preview: `python scripts/check_gui.py` with a working display, Tkinter and Pillow. It uses simulated data and performs no CAN I/O. `scripts/generate_schematic.py` rebuilds native sheets from `hardware/design/circuit.py`; PCB edits remain a separate CAD operation. `import_connector_models.py` is only for restoring/reviewing exact vendor CAD and transforms, followed by fresh STEP/render exports.

Compiler objects, downloaded Java dependencies, test runners, caches and scratch renderings are ignored. The retained firmware third-party subset supports STM32F042 and GCC; see its provenance. All checks are engineering evidence; physical qualification remains required.

Power transient simulation

The additional H power rerun was canceled; its report explicitly says INCOMPLETE. The LED/header checks are complete, but the full-release scripts below intentionally continue to require a complete power report. They will fail at that gate until a full run is requested and completed.

The optional native simulator is **ngspice 42 with KLU, XSPICE and PSpice compatibility**. Original manufacturer model files and hashes are retained in `sources/spice/`. The harness reads the maintained circuit's pin connections and component values; it generates simulator-specific copies, decks and raw traces under

ignored `tmp/spice/`. The full run includes nine cases plus a complete combined-startup solver repeat; the coupled switching models can take several hours.

```
# From current/, after source tools/env.sh; switching-model runs can take several hours.
python scripts/check_boost_spice.py --jobs 2
# Reuse only completed traces whose deck, model and simulator hashes match.
python scripts/check_boost_spice.py --jobs 2 --reuse
# Fast evidence consistency check (preserves and reports failed findings).
python scripts/check_boost_spice.py --verify-report
# Also require every screening criterion to pass; used by validate_release.sh.
python scripts/check_boost_spice.py --verify-report --require-pass
```

The published summary cover the protected boost stage, not the upstream buck or the whole board. A circuit, budget, source-model or harness change invalidates saved evidence. Each case has an eight-hour wall-time limit; a timeout or incomplete trace cannot pass. `--case NAME` produces separate partial reports and cannot replace the full release result. A failed screening criterion returns a nonzero exit status and remains visible in the report. See the simulation scope. Individual-case outputs remain under ignored `tmp/spice/`; the full run is required for the release report.

18 Sources and notices

Source: docs/SOURCES_AND_LICENSES.md

KiCad library/model notices are retained under `licenses/`, together with Arm CMSIS, ST device-header and WPILib notices. Firmware third-party directories and InteractiveHtmlBom output retain their notices. Original project work has no selected blanket license. Manufacturer documents/CAD remain their owners' material; retaining a file is not a redistribution-license grant.

Subject	Primary reference
Input fuse interruption / derating	Littelfuse 451/453
Buck and capacitors	TI LMR51420
Regulated 12 V boost	TI TPS61085, exact DGK VSSOP package
Shielded 3.3 µH inductor	Bourns SRP5030TA
Input/output/bias capacitor curve	Samsung CL32B106KBJNNW
Banner supply/residual voltage exceptions	QS18 product manual
Sensor 5 V latch-off limiter	TI TPS2553, -1 variant
Sensor 12 V output eFuse	TI TPS2596, TPS259620 latch-off variant
Comparator/leakage caveats	TI LM339LV
CAN interface / bypass	TI TCAN3413
STM32 pinout, supplies, clock	ST STM32F042
Example industrial output residual levels	Omron E3Z specifications
Terminal geometry and exact MPNs	Weidmüller 1824420000, 1824430000
CAN addressing / broadcast disable	WPILib

Samsung CL31A226KAHNNNE typical characteristic data is retained in `sources/datasheets/samsung_CL31A226_characteristics.json`, inherited from the earlier design's manufacturer retrieval. Typical curves do not establish guaranteed capacitance. DigiKey URLs, exact codes and dated observations are in `bom/catalog_china.json`. Vendor STEP URLs, hashes and transforms are in `hardware/library/3d/vendor_connectors.json`.

Equivalent parts need engineering review; distributor substitution suggestions do not establish electrical/mechanical equivalence. Availability and season rules can change after these observations.

Local source evidence includes the TI TPS61085/TPS2553 datasheets and Samsung CL31A226/CL32B106 characteristic JSON. Typical curves do not establish guaranteed minimum capacitance. U8's footprint and nominal STEP body are from the KiCad 10 Package_SO library. Horizontal connector STEP models are official vendor geometry. The new Samsung CL21B105/CL21B475 substitutions retain nominal ratings; the LDO derating allowance is analytical, not a measured characteristic curve.

Rev H sourcing URLs and tiers are in catalog_china.json. Header/cap geometry links the Harwin and Sullins manufacturer drawings; cost assumptions links the assembly fee and lead-time sources. Stock observations are not factory allocations.

Rev H input bias references: AO3401A, BSS138BK, BAT46W. The local SOT-23 footprint/model comes from installed KiCad libraries under their existing library-license policy. CTRE input-mode API informed feature choices; no CTRE firmware or hardware design is included.

Power transient models: TI TPS61085 SLIM133B, TI TPS2553-1 SLVM426A, TI TPS259620 SLVMD54A, and Diodes B240A. Local provenance and hashes distinguish original archive bytes from the diode model's plaintext transcription. Manufacturer notices remain in each file. They are manufacturer material, not covered by a project license. The harness only creates temporary compatibility adaptations: unique helper names, duplicate `PARAMS:` cleanup, and PSpice Boolean XOR conversion. It uses Gear integration after observing an artificial latch reset with trapezoidal integration. Simulator compatibility is described by the ngspice manual.

Rev H LED/header: OPSCO SK6805-EC15 A/1, TI SN74AHCT1G125, XFCN PM254V-12-10P-H85. Local geometry provenance describes the project-created nominal LED and female-socket envelopes. ST UM2910 defines the programmer mapping. ST RM0091, DMA table 29, assigns TIM2_UP to DMA1 channel 2; STM32F042 AF2 maps PB3 to TIM2_CH2.